MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE-JUNIOR RECREATIONAL BUILDING
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Junior Recreational Building in Village Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Junior Recreational Building only.

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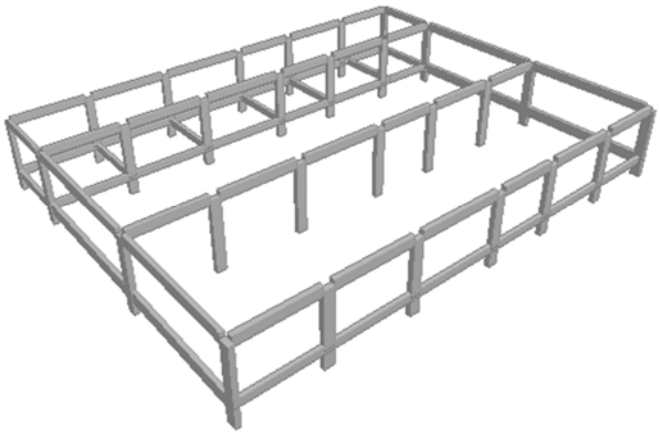
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-300 Topped) b/w Grid 2 & 3	= **3.88 KN/m2**
(HCS-150 Topped) b/w Grid 1 & 2	= **2.52 KN/m2**
(HCS-250 Topped) b/w Grid 3 & 4	= **3.43 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load b/w Grid 2 & 3 = **7.18 kN/m²**

Total Load b/w Grid 1 & 2 = **5.82 kN/m²**

Total Load b/w Grid 3 & 4 = **6.73 kN/m²**

UDL on External Beams along Grid Line-4 = 9.05 x 6.73/2 =**30.45 Kn/m**

UDL on Internal Beam along Grid Line-2 = 14.40+ (10.05*7.18/2) =**50.47 Kn/m**

UDL on Internal Beam along Grid Line-3 = 30.45+ (10.05*7.18/2) =**66.52 Kn/m**

UDL on External Beams along Grid Line-1 = 4.95*5.82/2 =**14.40 Kn/m**

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Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU

16 x 0.25 x 3.0m clear height = **12.0 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat = **0.50 kN/m²**

UDL on External Beams along Grid Line-4 = 9.05 x 0.50/2 =**2.26 Kn/m**

UDL on Internal Beam along Grid Line-2 = 1.24+ (10.05*0.50/2) =**3.75 Kn/m**

UDL on Internal Beam along Grid Line-3 = 2.26+ (10.05*0.50/2) =**4.77 Kn/m**

UDL on External Beams along Grid Line-1 = 4.95*0.50/2 =**1.24 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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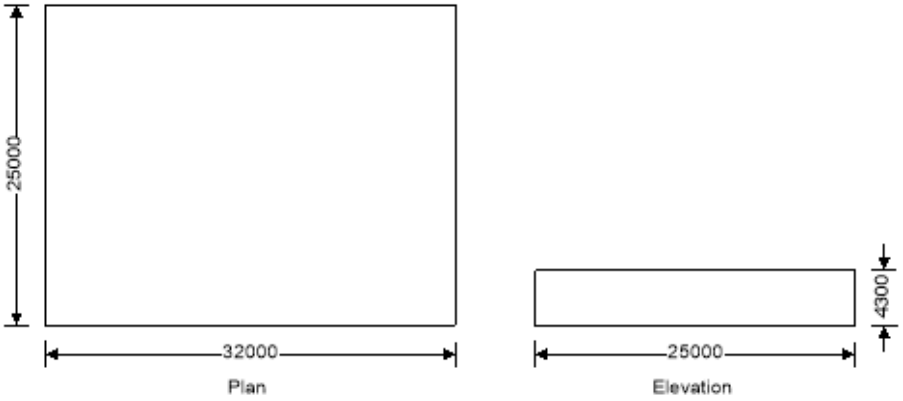
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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 32000 mm
Width of building;	d = 25000mm
Height to eaves;	H = 4300 mm
Mean height;	h = 4300 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)

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Gust effect factor; G_f = 0.85(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3600	1.03	1.29
3600	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A (-ve)	4.30	-0.90	1.29	-1.69	68.80	-116.58
B (-ve)	4.30	-0.90	1.29	-1.69	68.80	-116.58
C (-ve)	4.30	-0.50	1.29	-1.26	137.60	-172.87
D (-ve)	4.30	-0.30	1.29	-1.04	524.80	-544.38

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A (+ve)	4.30	-0.18	1.29	0.51	68.80	35.20
B (+ve)	4.30	-0.18	1.29	0.51	68.80	35.20
C (+ve)	4.30	-0.18	1.29	0.51	137.60	70.39
D (+ve)	4.30	-0.18	1.29	0.51	524.80	268.47

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.30	0.80	1.29	0.17	107.52	18.01
B	4.30	-0.44	1.29	-1.20	107.50	-128.46
C	4.30	-0.70	1.29	-1.48	137.60	-203.02
D	4.30	-0.70	1.29	-1.48	137.60	-203.02

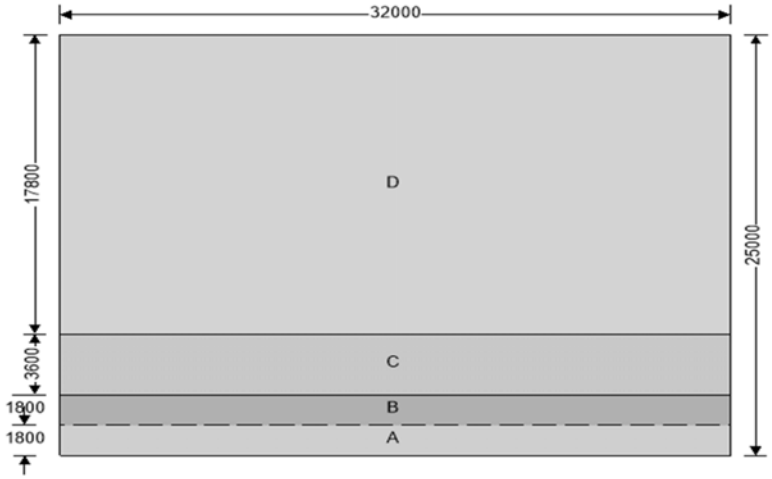
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

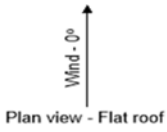
Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	4.30	0.80	1.29	1.58	107.52	170.41
B	4.30	-0.44	1.29	0.22	107.50	23.91
C	4.30	-0.70	1.29	-0.06	137.60	-7.98
D	4.30	-0.70	1.29	-0.06	137.60	-7.98

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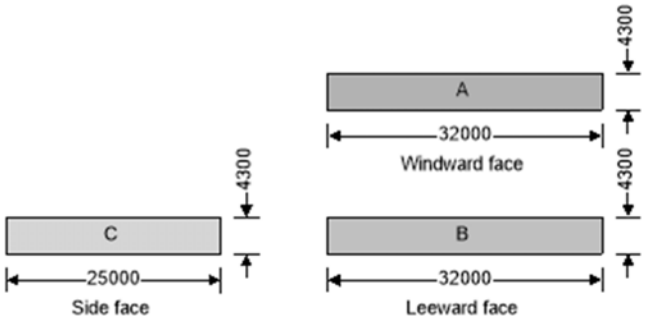
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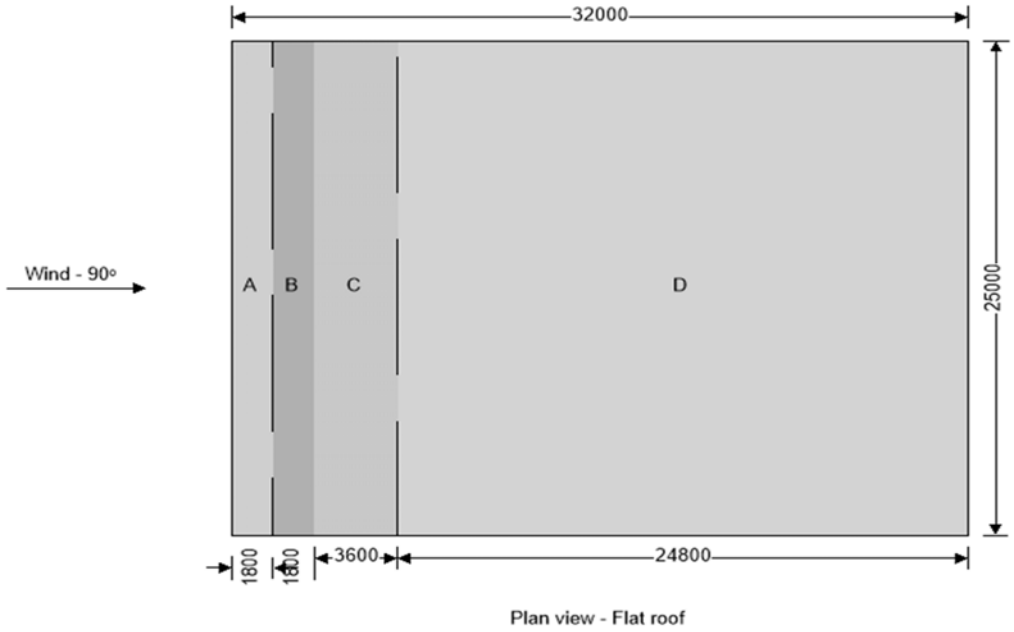
Plan view - Flat roof



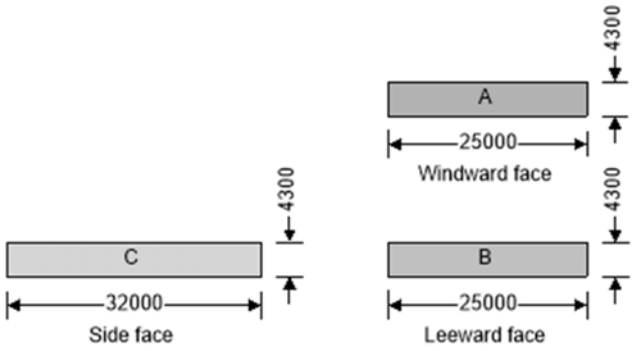
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Plan view - Flat roof



So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*S1 = 2/3 * 1.60 * 0.10 = 0.10 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 second

S_{DS} Damped Design Spectral Response Acceleration@short period

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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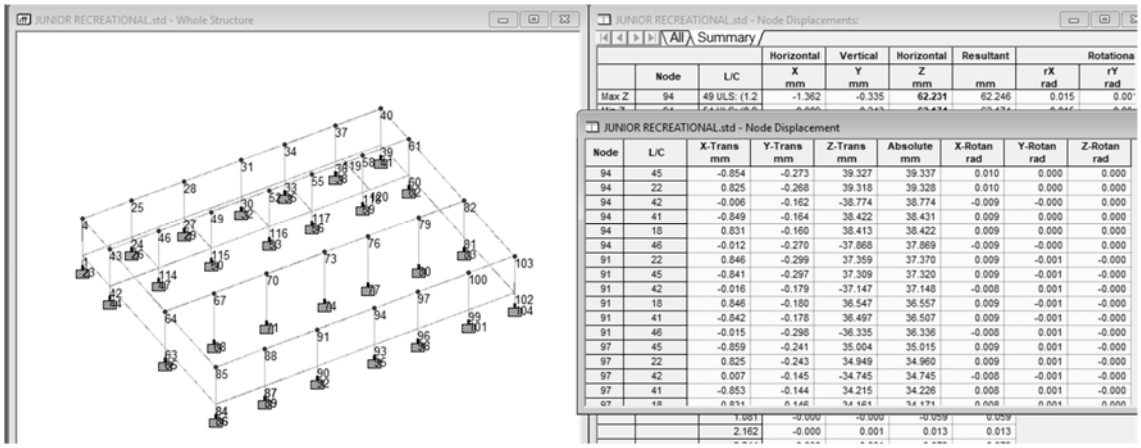
8. DEFLECTIONS/DRIFT CHECK

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Node	L/C	Horizontal X mm	Vertical Y mm	Horizontal Z mm	Resultant mm	Rotationa rX rad	rY rad
Max Z	94	49 ULS: (1.2	-1.362	-0.335	62.231	62.246	0.015

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
94	45	-0.854	-0.273	39.327	39.337	0.010	0.000	0.000
94	22	0.825	-0.268	39.318	39.328	0.010	0.000	0.000
94	42	-0.006	-0.162	-38.774	38.774	-0.009	-0.000	0.000
94	41	-0.849	-0.164	38.422	38.431	0.009	0.000	0.000
94	18	0.831	-0.160	38.413	38.422	0.009	0.000	0.000
94	46	-0.012	-0.270	-37.868	37.869	-0.009	-0.000	0.000
91	22	0.846	-0.299	37.359	37.370	0.009	-0.001	-0.000
91	45	-0.841	-0.297	37.309	37.320	0.009	-0.001	-0.000
91	42	-0.016	-0.179	-37.147	37.148	-0.008	0.001	-0.000
91	18	0.846	-0.180	36.547	36.557	0.009	-0.001	-0.000
91	41	-0.842	-0.178	36.497	36.507	0.009	-0.001	-0.000
91	46	-0.015	-0.298	-36.335	36.336	-0.008	0.001	-0.000
97	45	-0.859	-0.241	35.004	35.015	0.009	0.001	-0.000
97	22	0.825	-0.243	34.949	34.960	0.009	0.001	-0.000
97	42	0.007	-0.145	-34.745	34.745	-0.008	-0.001	-0.000
97	41	-0.853	-0.144	34.215	34.226	0.008	0.001	-0.000

Maximum attained Drift as per Analysis from above figure = 39 mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 4300/100 = 43mm

Hence the Structure is safe considering drift check…

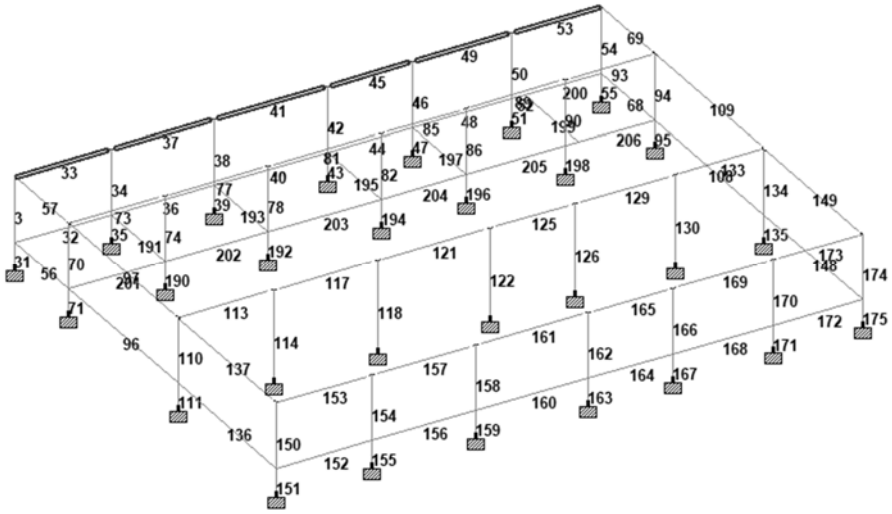
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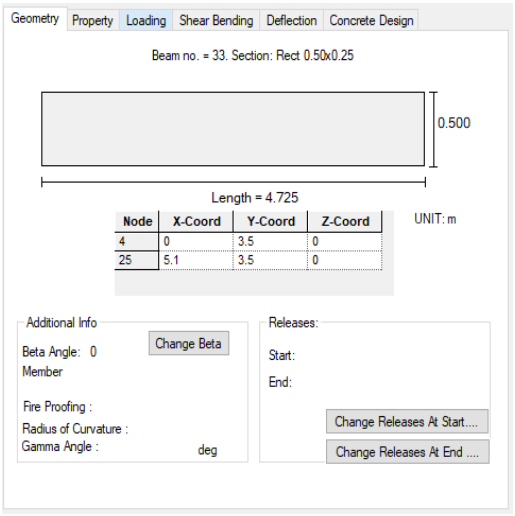
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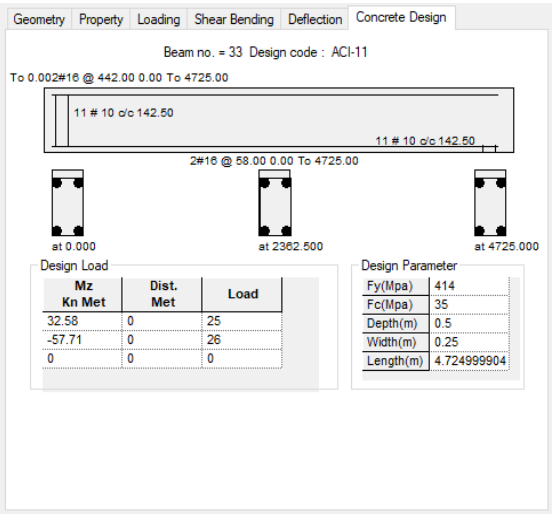
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **57 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm^2**

Maximum Moment required from Analysis = **32 Knm (Sagging)**

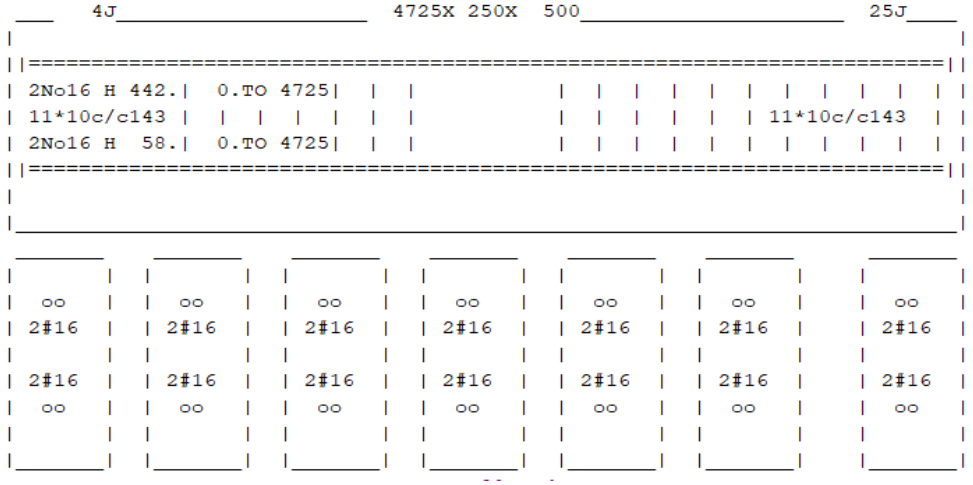
However Provide minimum 3-16 dia -Full Straight Bars Top & 3-20 dia-Full Straight Bottom.

9.1.1. Staad Output For LB1

```
            B E A M   N O.    33 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   22.47 KNS  Vc=  106.71 KNS  Vs=    0.00 KNS
 Tu=   15.02 KN-MET  Tc=    4.0 KN-MET  Ts=   20.0 KN-MET  LOAD    29
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1422. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.95 SQ.CM.

 AT END   SUPPORT - Vu=   37.65 KNS  Vc=  119.13 KNS  Vs=    0.00 KNS
 Tu=   13.09 KN-MET  Tc=    3.9 KN-MET  Ts=   17.5 KN-MET  LOAD    28
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1422. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.31 SQ.CM.
```

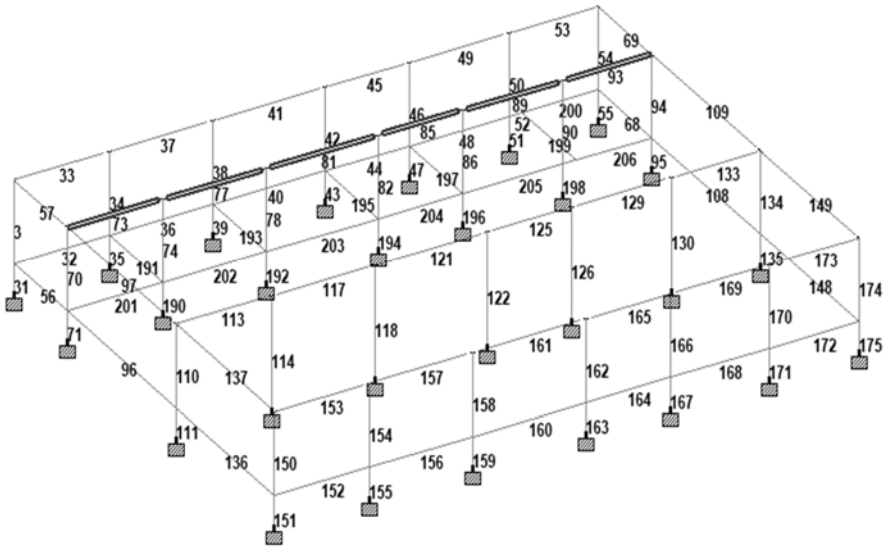


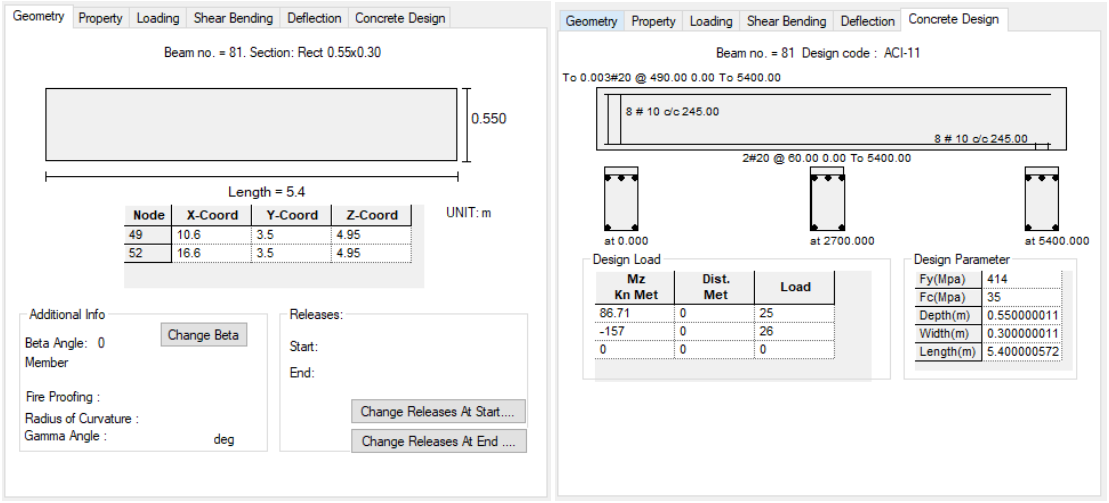
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9.2. Design of Ledger Beam(LB2)





Design of Ledger Beam (LB2)

Provide 3-20 dia-Full Straight Bars Bottom & 3-20 dia-Full Straight Top considering Long term deflection check as shown below in section 9.2.1.

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9.2.1. Long Term Deflection Check for LB2 Beam:

INPUT DATA:

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	550	mm
width of beam, b	=	300	mm
Length of beam, L	=	6.00	m
Bottom bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of bottom bars	=	3.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff\ bot}$ (@ bottom), d	=	490.00	mm
Mdead	=	58.00	kN-m
M live+temp	=	26.00	kN-m

(FROM STAAD RESULTS)

CHECK DEFLECTION: $M_{d+\ell}$ = 84.00 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 71.00 kN-m

modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa *?ACI(318M-08) Sec. 9.5.2.3*

$n_s = \frac{Es}{Ec}$ = 7.19

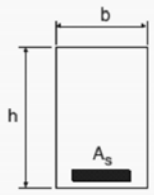
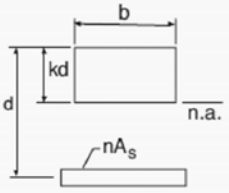
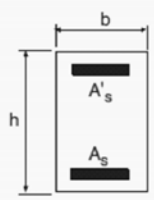
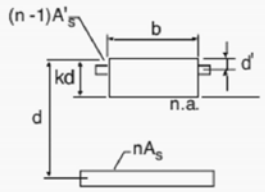
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 4.16E+09 \text{ mm?}$$

$$B = \frac{b}{nAs} = 0.04 \text{ mm/mm²}$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 127.92 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.10E+09 \text{ mm?}$$

$$\frac{I_g}{I_{cr}} = 3.79$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 55.48 \text{ kN-m} \qquad \text{?ACI(318M-08) Sec. 9.5.2.3}$$

under dead load

$$\frac{M_{cr}}{M_d} = 0.96$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 3.78E+09 \text{ mm?} \qquad \text{?ACI(318M-08) Sec. 9.5.2.3}$$

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.78$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 2.56E+09 \text{ mm?}$$?ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.66$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 1.98E+09 \text{ mm?}$$?ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 2.071 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 3.743 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 5.722 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 3.651 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\ell \le \frac{\ell}{180} = 33.33 \text{ mm} > 3.65 \text{ mm} \quad ok!$$

$$(\Delta_i)_\ell \le \frac{\ell}{360} = 16.67 \text{ mm} > 3.65 \text{ mm} \quad ok!$$

additional long-term deflection

$$\xi = 2.00$$

5 years or more	2.0
12 months	1.4
6 months	1.2
3 months	1.0

$$\rho = \frac{As}{bd} = 0.0064$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.51$$

?ACI(318M-08) Sec. 9.5.2.5

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 5.67 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell = 9.32 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480} = 12.50 \text{ mm} > 9.32 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240} = 25.00 \text{ mm} > 9.32 \text{ mm} \quad ok!$$

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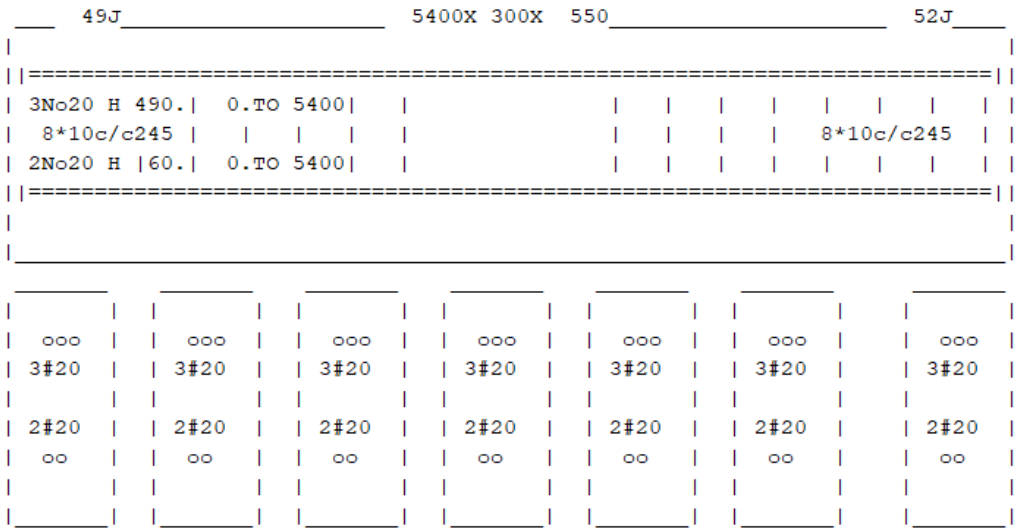
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9.2.2. Staad Output For LB2

```
                B E A M  N O.    81 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=    82.13 KNS   Vc=  143.10 KNS   Vs=     0.00 KNS
Tu=     0.03 KN-MET   Tc=     5.9 KN-MET   Ts=     0.0 KN-MET   LOAD     17
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 1660. MM

AT END    SUPPORT - Vu=    77.59 KNS   Vc=  143.10 KNS   Vs=     0.00 KNS
Tu=     0.03 KN-MET   Tc=     5.9 KN-MET   Ts=     0.0 KN-MET   LOAD     17
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 245. MM  C/C FOR 1660. MM
```

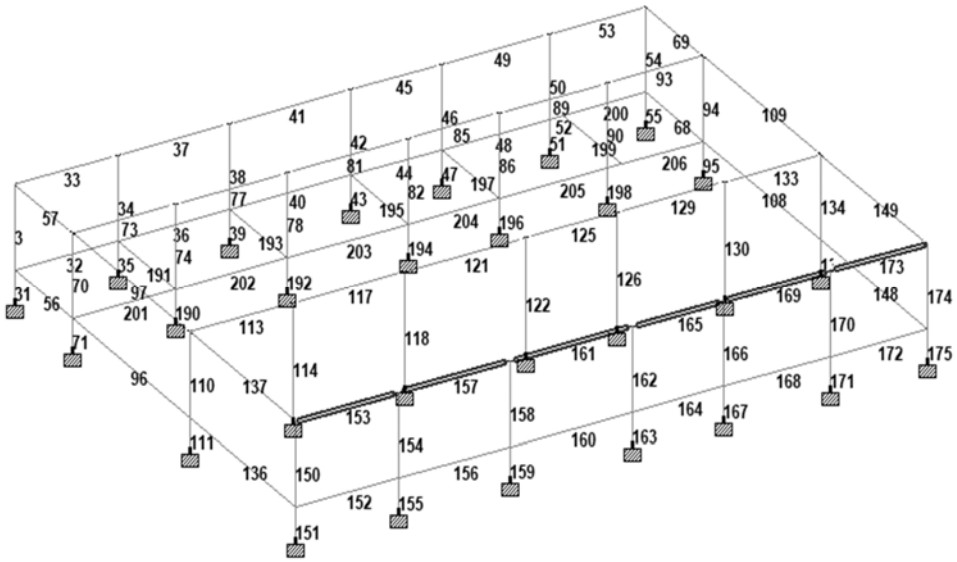


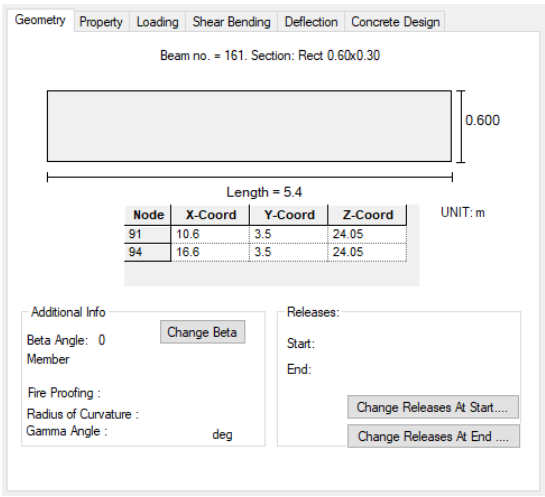
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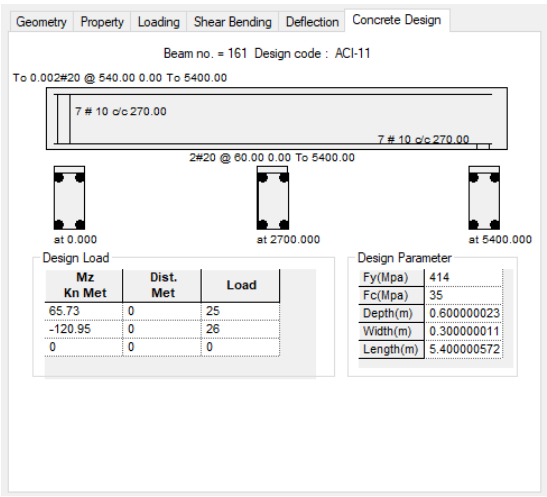
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9.3. Design of Ledger Beam(LB3)







Hence Provide 3-20 dia-Full Straight Bars Top & Bottom considering minimum reinfrocement requirements…

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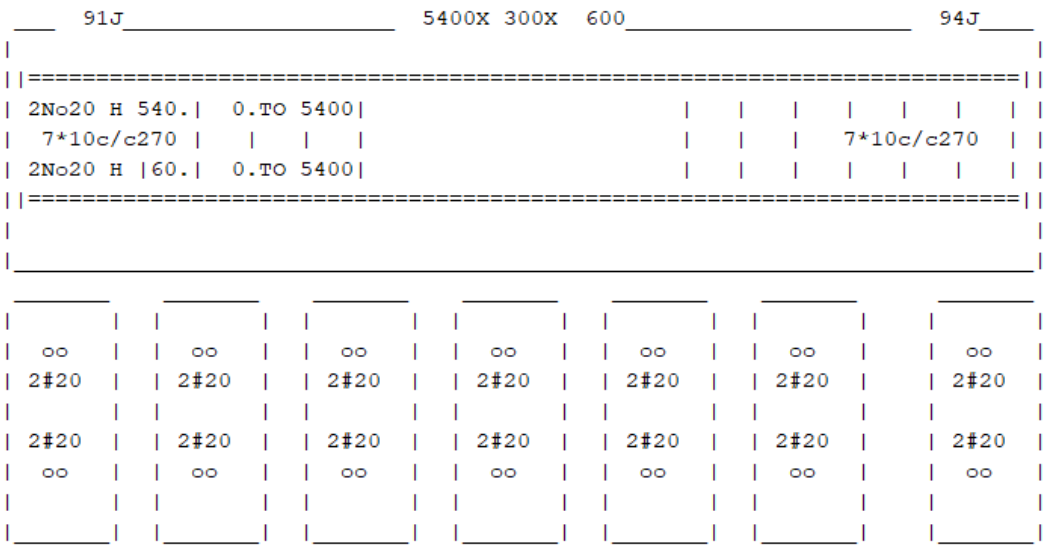
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9.3.1. Staad Output For LB3

```
                B E A M  N O.   161 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=   68.47 KNS  Vc=  158.89 KNS  Vs=    0.00 KNS
  Tu=    2.94 KN-MET  Tc=    6.6 KN-MET  Ts=    0.0 KN-MET  LOAD    28
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 270. MM  C/C FOR 1560. MM

  AT END   SUPPORT - Vu=   64.97 KNS  Vc=  158.89 KNS  Vs=    0.00 KNS
  Tu=    1.09 KN-MET  Tc=    6.6 KN-MET  Ts=    0.0 KN-MET  LOAD    28
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 270. MM  C/C FOR 1560. MM
```

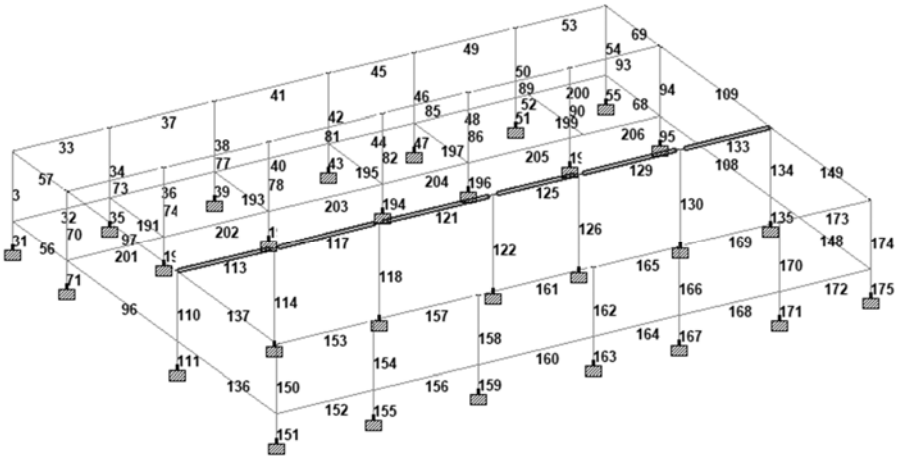


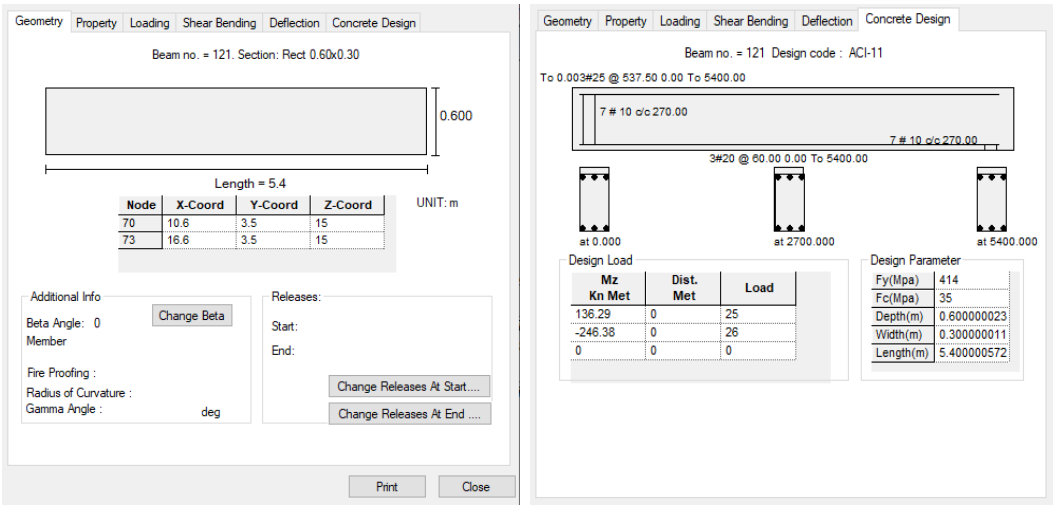
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9.4. Design of Ledger Beam(LB4)





Design of Ledger Beam (LB4)

Provide 5-20 dia-Full Straight Bars Bottom & 3-25 dia-Full Straight Top considering Long term deflection check as shown below in section 9.4.1.

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9.4.1. Long Term Deflection Check for LB4 Beam:

INPUT DATA:

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	600	mm
width of beam, b	=	300	mm
Length of beam, L	=	6.00	m
Bottom bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of bottom bars	=	5.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff\ bot}$ (@ bottom), d	=	540.00	mm
Mdead	=	95.00	kN-m
M live+temp	=	30.00	kN-m

(FROM STAAD RESULTS)

CHECK DEFLECTION: $M_{d+\ell}$ = 125.00 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 110.00 kN-m

modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa *?ACI(318M-08) Sec. 9.5.2.3*

$n_s = \frac{Es}{Ec}$ = 7.19

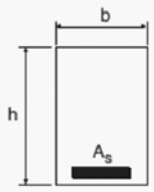
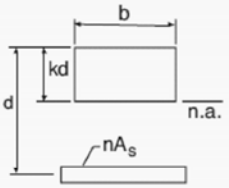
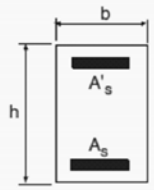
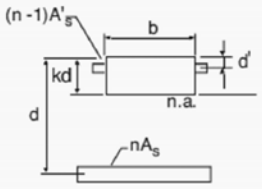
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 5.40E+09 \text{ mm?}$$

$$B = \frac{b}{nAs} = 0.03 \text{ mm/mm²}$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 167.50 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 2.04E+09 \text{ mm?}$$

$$\frac{I_g}{I_{cr}} = 2.65$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 66.02 \text{ kN-m}$$ *?ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.69$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 3.17E+09 \text{ mm?}$$ *?ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.60$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 2.76E+09 \text{ mm?}$$?ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.53$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 2.53E+09 \text{ mm?}$$?ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 4.046 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 5.366 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 6.655 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 2.609 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\ell \le \frac{\ell}{180} = 33.33 \text{ mm} > 2.61 \text{ mm} \quad ok!$$

$$(\Delta_i)_\ell \le \frac{\ell}{360} = 16.67 \text{ mm} > 2.61 \text{ mm} \quad ok!$$

additional long-term deflection

$\xi = 2.00$

5 years or more	2.0
12 months	1.4
6 months	1.2
3 months	1.0

$$\rho = \frac{As}{bd} = 0.0097$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.35$$

?ACI(318M-08) Sec. 9.5.2.5

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 7.23 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell = 9.84 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480} = 12.50 \text{ mm} > 9.84 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240} = 25.00 \text{ mm} > 9.84 \text{ mm} \quad ok!$$

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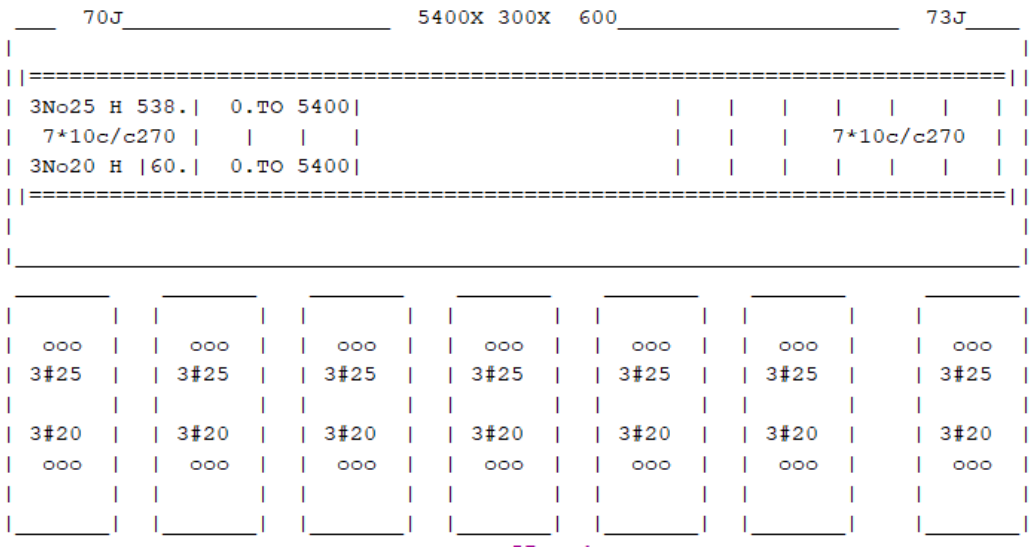
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9.4.2. Staad Output For LB4

```
                B E A M  N O.   121 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=  113.72 KNS  Vc=  282.30 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 270. MM  C/C FOR 1560. MM

AT END   SUPPORT - Vu=  107.09 KNS  Vc=  234.07 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 270. MM  C/C FOR 1560. MM
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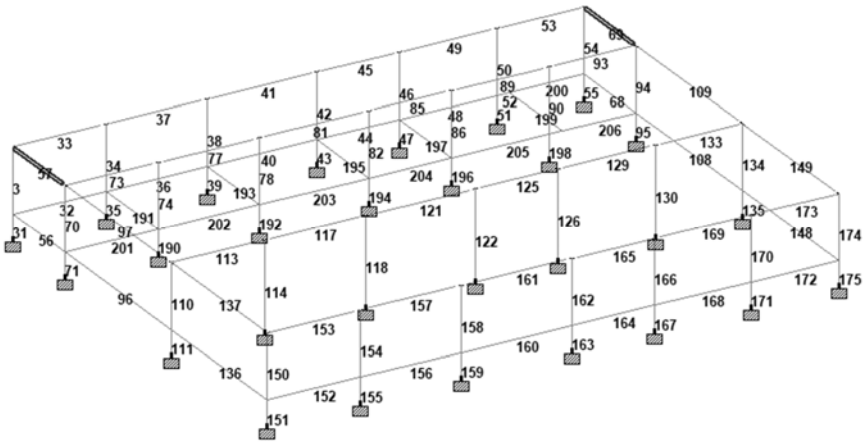


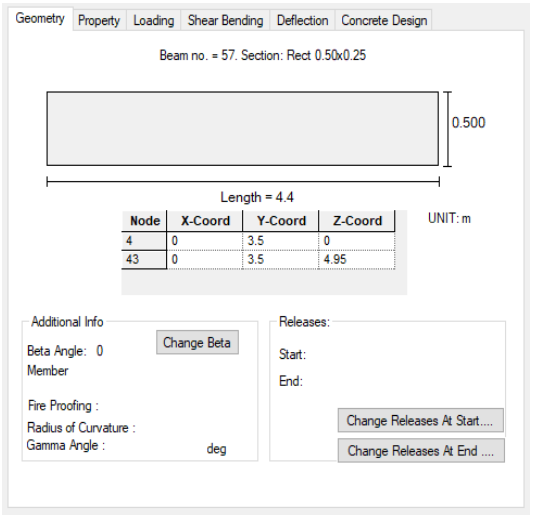
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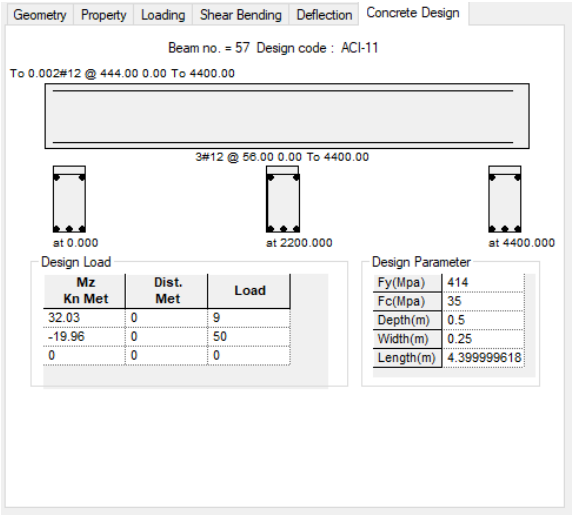
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9.5. Design of Beam-(B1)







Design of Beam (B1)

However Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

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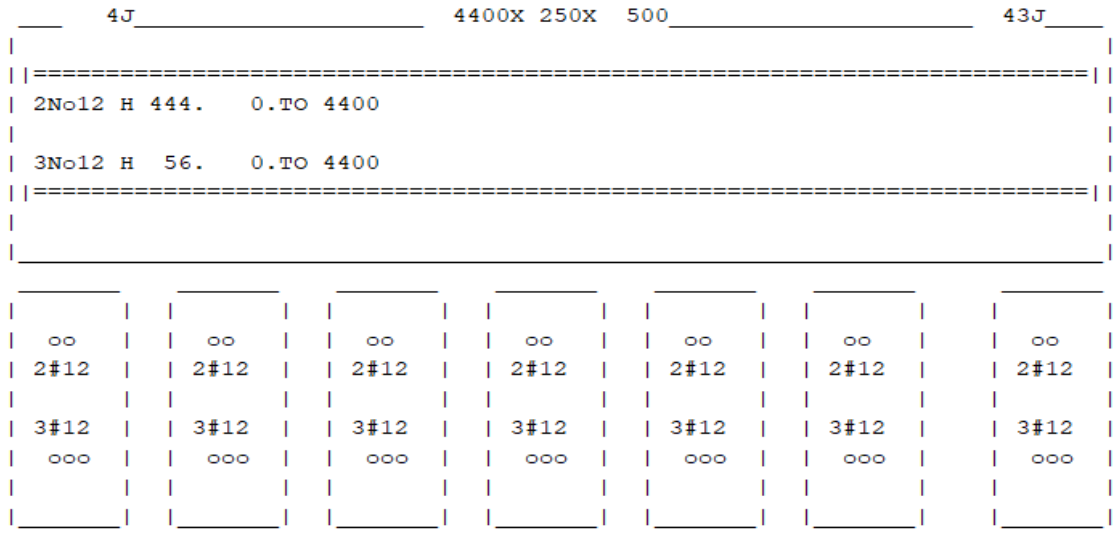
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9.5.1. Staad Output For B1

```
                B E A M  N O.    57 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=     3.93 KNS   Vc=  147.42 KNS   Vs=     0.00 KNS
   Tu=     0.16 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=     3.51 KNS   Vc=  191.79 KNS   Vs=     0.00 KNS
   Tu=     2.36 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
```

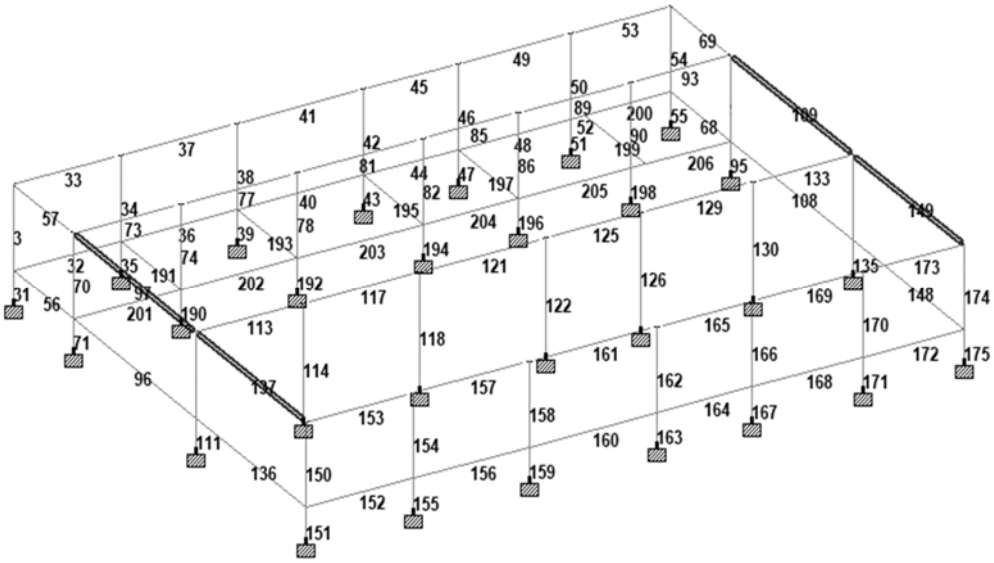


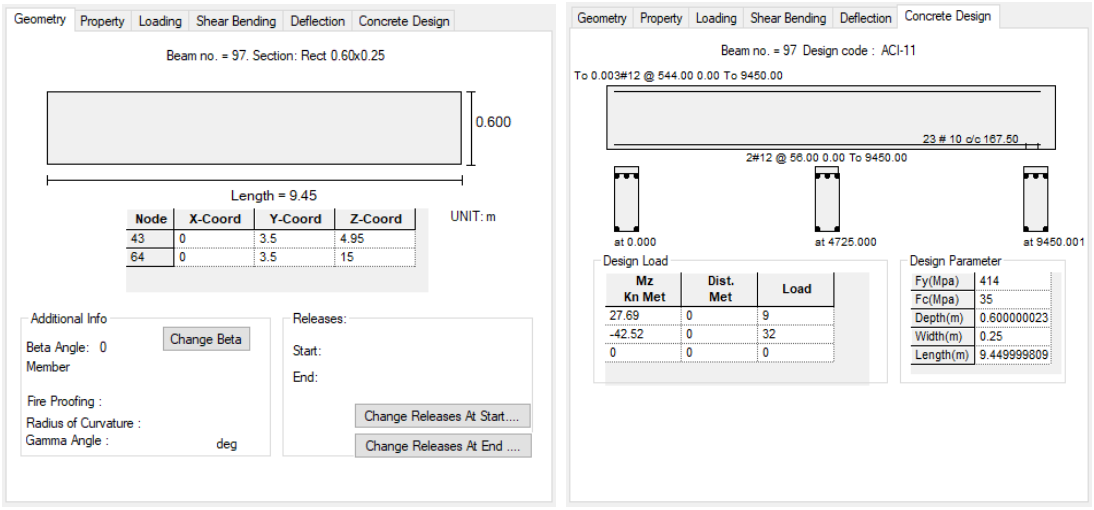
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9.6. Design of Beam-(B2)





Design of Beam (B2)

However Provide minimum 3-20 dia-Full Straight Bars Top & Bottom.

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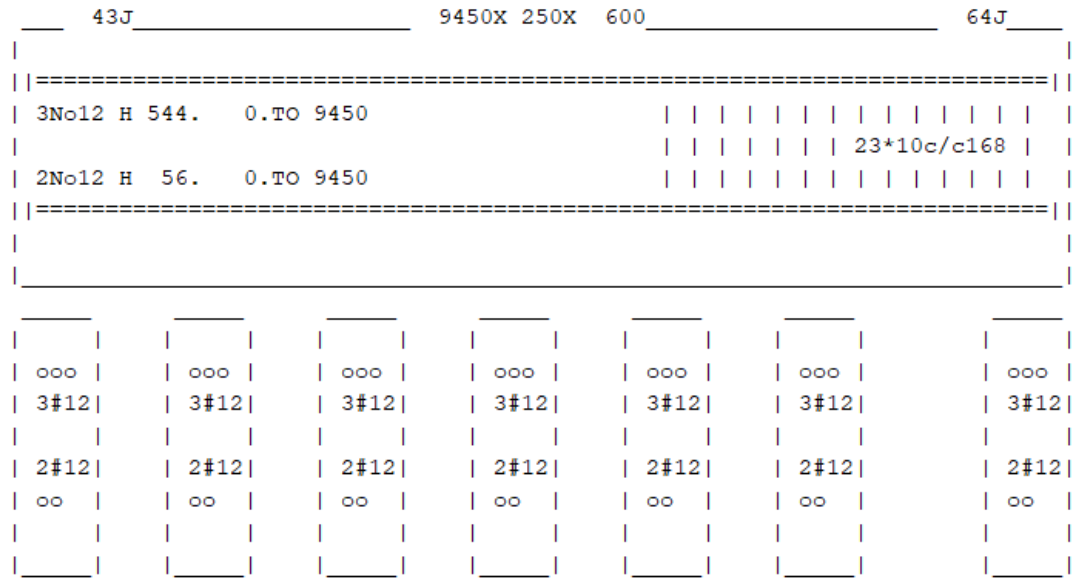
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9.6.1. Staad Output For B2

```
        B E A M  N O.    97 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   12.65 KNS  Vc=  131.61 KNS  Vs=    0.00 KNS
 Tu=    3.06 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   17.85 KNS  Vc=  131.28 KNS  Vs=    0.00 KNS
 Tu=    6.05 KN-MET  Tc=    5.1 KN-MET  Ts=    8.1 KN-MET  LOAD    25
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 168. MM  C/C FOR 3585. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.88 SQ.CM.
```



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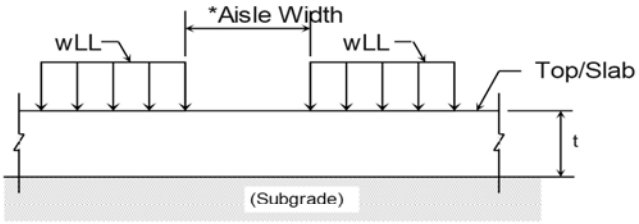
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm	
Concrete Strength, f 'c =	5000	psi	35.000	kpa	
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)
Factor of Safety, FS =	1.500		1.500		
Uniform Live Load, wLL =	104.45	psf	5.00	kpa	



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-6mm Dia@200 c-c Both Ways-TOP

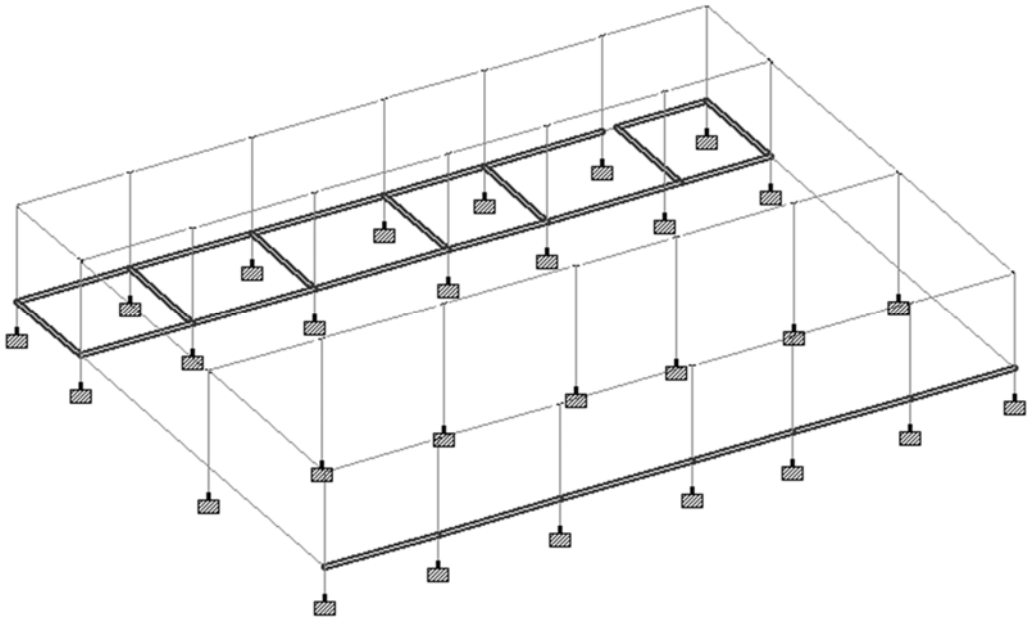
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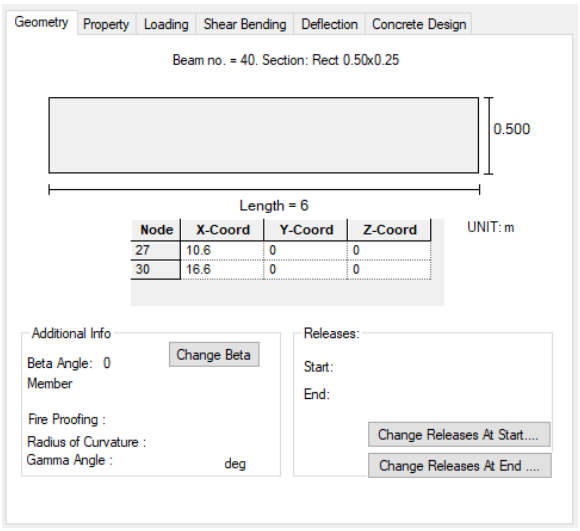
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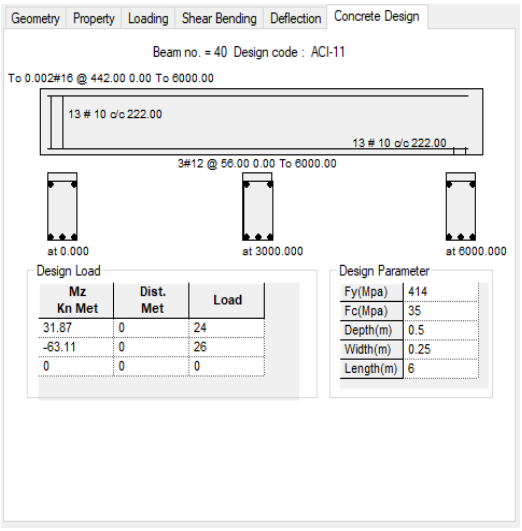
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11. DESIGN OF BEAM-(GB1)

11.1. DESIGN OF BEAM-(GB1)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

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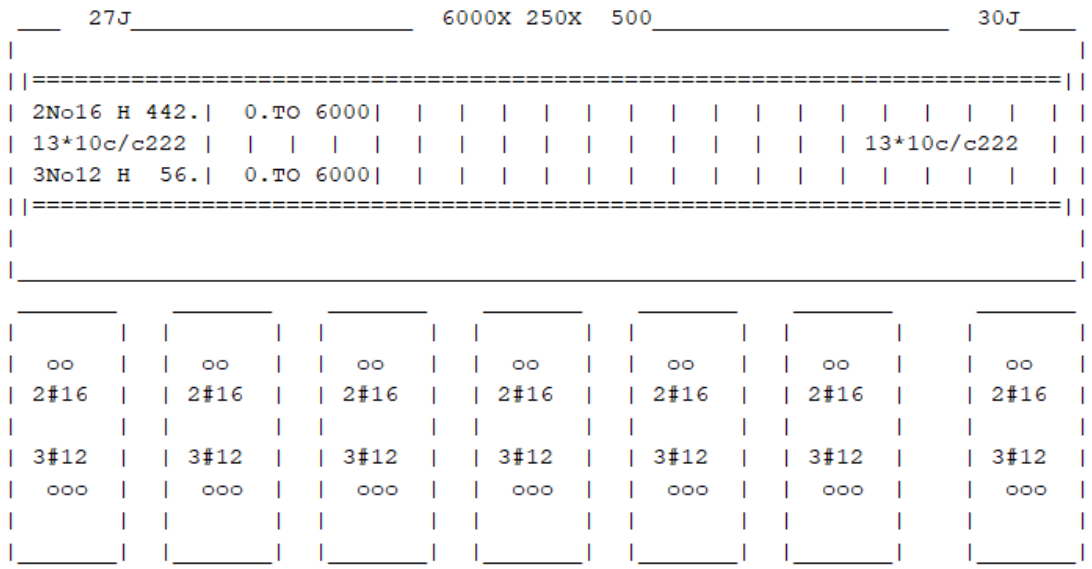
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11.1.1. STAAD OUTPUT FOR GB1

```
            B E A M   N O.    40 D E S I G N   R E S U L T S - SHEAR


 AT START SUPPORT - Vu=    53.85 KNS   Vc=  106.56 KNS   Vs=     0.00 KNS
 Tu=     0.00 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD     24
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 2560. MM

 AT END    SUPPORT - Vu=    53.33 KNS   Vc=  106.64 KNS   Vs=     0.00 KNS
 Tu=     0.00 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD     24
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 222. MM  C/C FOR 2560. MM
```

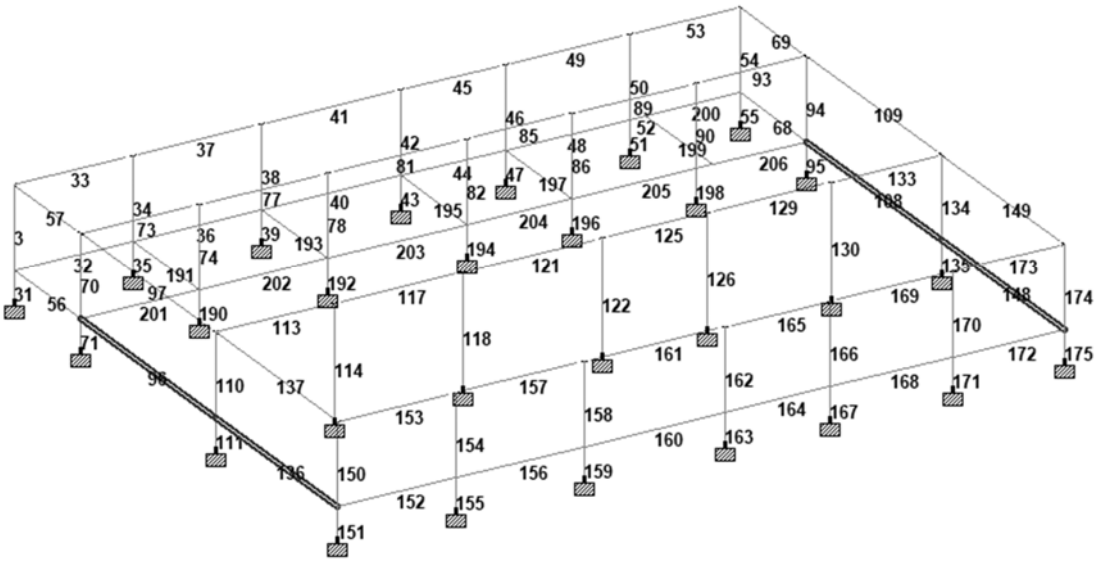


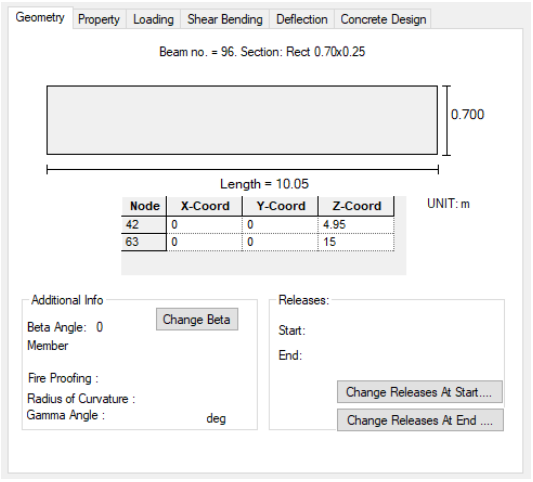
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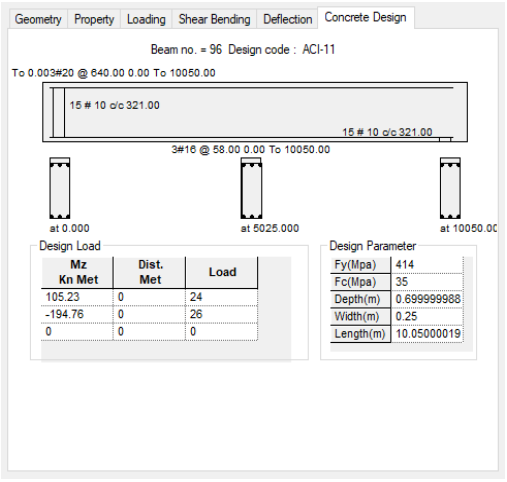
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11.2. DESIGN OF BEAM-(GB2)







Provide minimum 3-20 dia-Full Straight Bars Top & Bottom.

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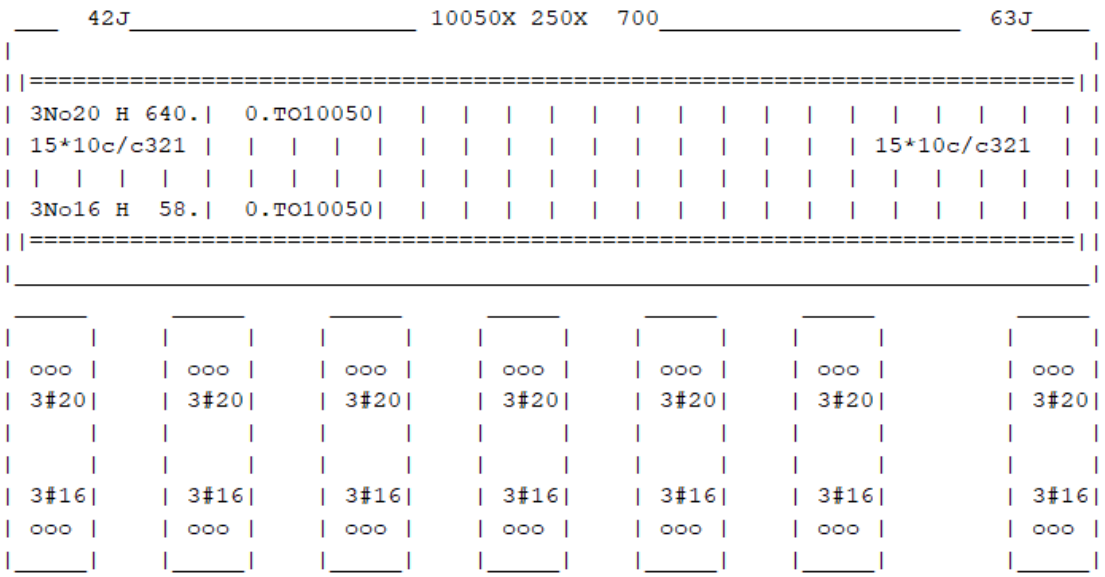
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11.2.1. STAAD OUTPUT FOR GB2

```
                B E A M  N O.    96 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=   68.64 KNS  Vc=  156.92 KNS  Vs=    0.00 KNS
 Tu=    0.50 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD     3
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 321. MM  C/C FOR 4385. MM

 AT END   SUPPORT - Vu=   72.85 KNS  Vc=  155.60 KNS  Vs=    0.00 KNS
 Tu=    0.50 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD     3
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 321. MM  C/C FOR 4385. MM
```

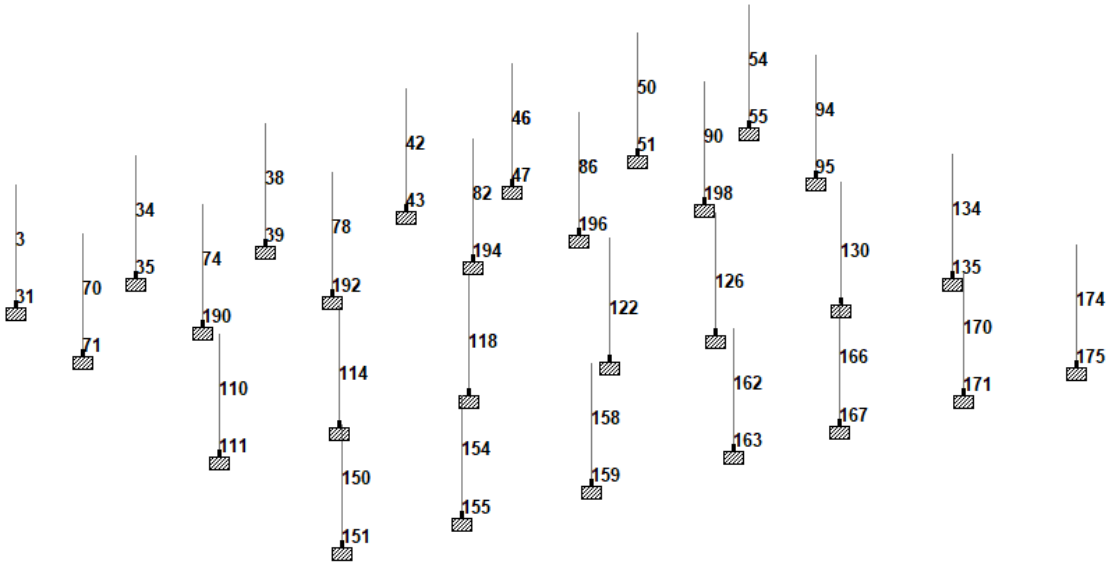


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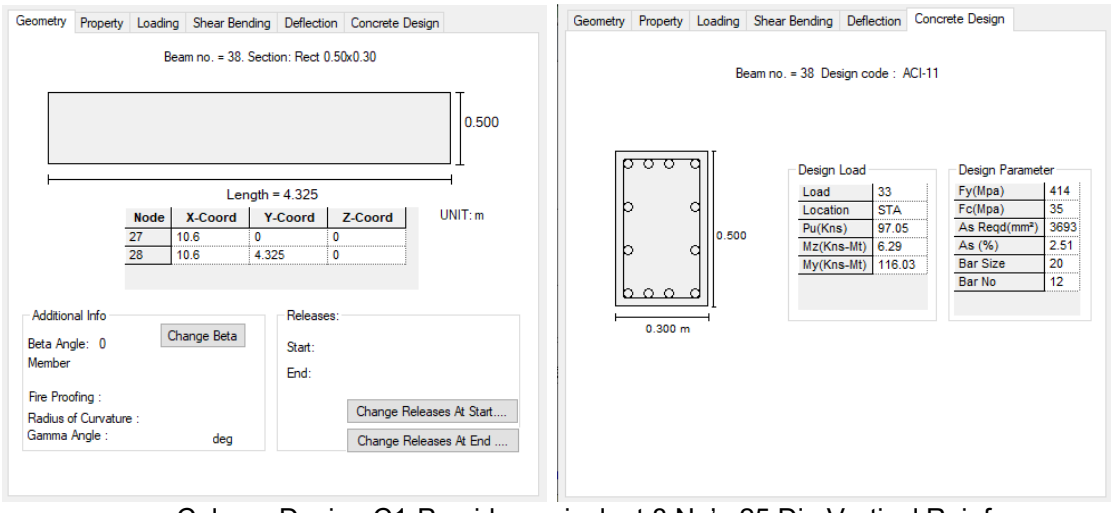
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12. CONCRETE COLUMN DESIGN



Column numbers

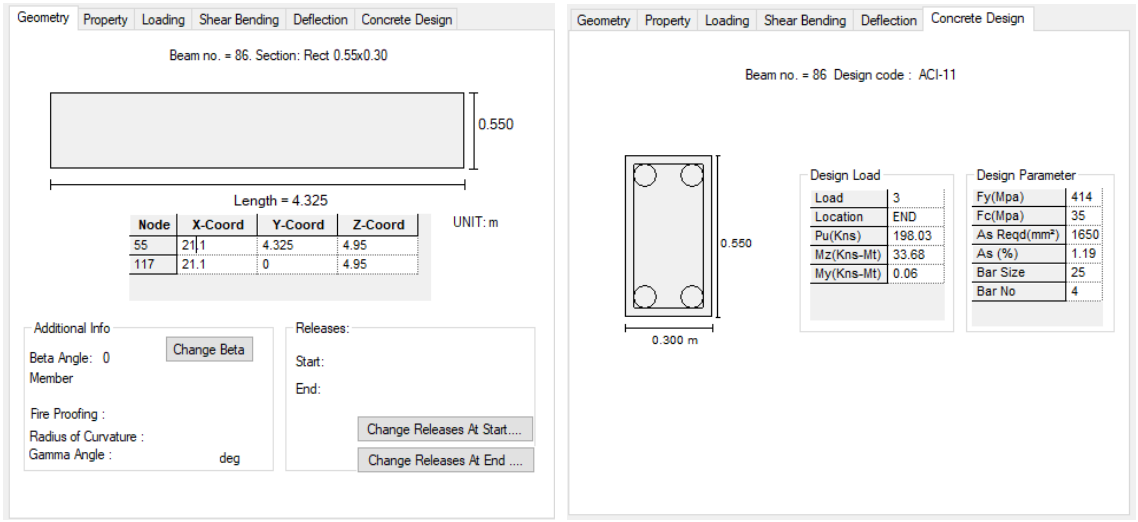


Column Design C1-Provide equivalent 8 No's-25 Dia Vertical Reinf.

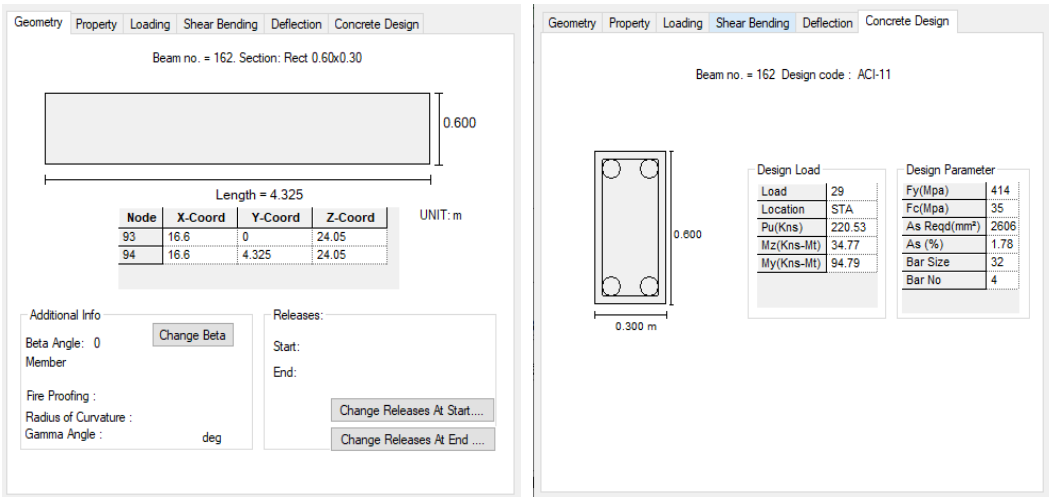
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Column Design C2-Provide equivalent 8 No's-20 Dia Vertical Reinf.

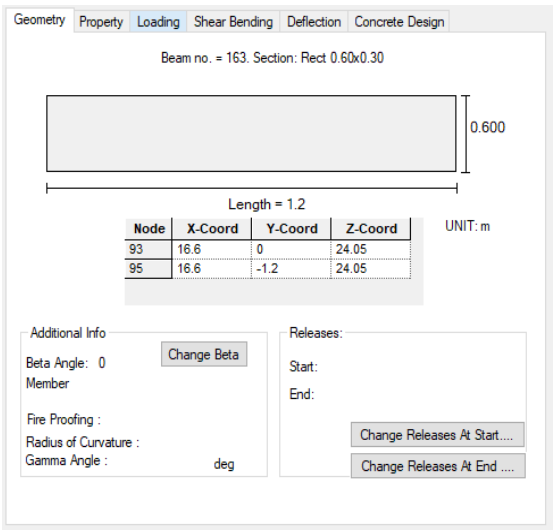


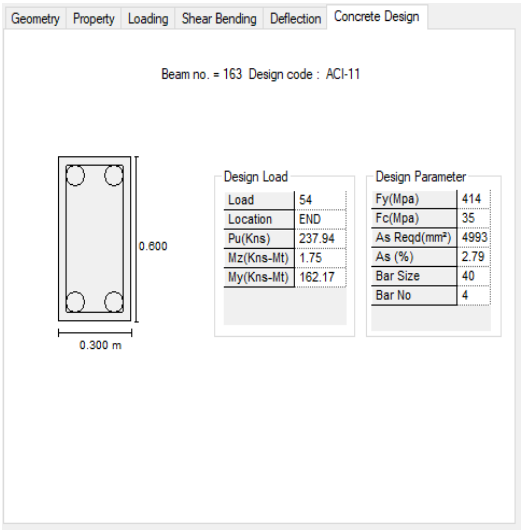
Column Design C3-Provide equivalent 8 No's-25 Dia Vertical Reinf.

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Column Design CR-Provide equivalent 10 No's-25 Dia Vertical Reinf.

12.1. Staad Output For Column C1, C2, C3 & CR

```
ACI 318-11   COLUMN  NO.    38   DESIGN RESULTS
==================================================

FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 300.0 X 500.0 MMS, TIED
       AREA OF STEEL REQUIRED =  3693.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

 12 - 20 MM               2.513       53      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM
```


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ACI 318-11 COLUMN NO. 86 DESIGN RESULTS

----------< PAGE 16 Ends Here >----------

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 300.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1650.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.190	3	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 300.00 MM

ACI 318-11 COLUMN NO. 162 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 300.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 2606.4 SQ. MM

----------< PAGE 23 Ends Here >----------

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 32 MM	1.787	29	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 300.00 MM

ACI 318-11 COLUMN NO. 163 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 300.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 4993.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 40 MM	2.793	54	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 300.00 MM

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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b I_e/360 **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = $M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-1800-EG-AR-GAD-3135
2.)	OVERALL LAYOUT PLAN	MD-216-1800-EG-CV-GAD-1535
3.)	LAYOUT OF ROOF & GRADE SLAB	MD-216-1800-EG-CV-GAD-1514
4.)	LAYOUT OF FDN, COL & GRADE BEAM	MD-216-1800-EG-CV-GAD-1501
5.)	DETAILS OF ROOF & GRADE SLAB	MD-216-1800-EG-CV-GAD-1515
6.)	DETAILS OF FDN, COL & GRADE BEAM	MD-216-1800-EG-CV-GAD-1502
7.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
8.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001
9.)	BUILDING ELEVATIONS & SECTIONS	MD-216-1800-EG-AR-GAD-3136

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 4 0 4.325 0; 23 0 -1.2 0; 24 5.1 0 0; 25 5.1 4.325 0; 26 5.1 -1.2 0;
27 10.6 0 0; 28 10.6 4.325 0; 29 10.6 -1.2 0; 30 16.6 0 0; 31 16.6 4.325 0;
32 16.6 -1.2 0; 33 21.1 0 0; 34 21.1 4.325 0; 35 21.1 -1.2 0; 36 26.4 0 0;
37 26.4 4.325 0; 38 26.4 -1.2 0; 39 31.15 0 0; 40 31.15 4.325 0;
41 31.15 -1.2 0; 42 0 0 4.95; 43 0 4.325 4.95; 44 0 -1.2 4.95;
46 5.1 4.325 4.95; 47 5.1 -1.2 4.95; 49 10.6 4.325 4.95; 50 10.6 -1.2 4.95;
52 16.6 4.325 4.95; 53 16.6 -1.2 4.95; 55 21.1 4.325 4.95; 56 21.1 -1.2 4.95;
58 26.4 4.325 4.95; 59 26.4 -1.2 4.95; 60 31.15 0 4.95; 61 31.15 4.325 4.95;
62 31.15 -1.2 4.95; 63 0 0 15; 64 0 4.325 15; 65 0 -1.2 15; 67 5.1 4.325 15;
68 5.1 -1.2 15; 70 10.6 4.325 15; 71 10.6 -1.2 15; 73 16.6 4.325 15;
74 16.6 -1.2 15; 76 21.1 4.325 15; 77 21.1 -1.2 15; 79 26.4 4.325 15;
80 26.4 -1.2 15; 81 31.15 0 15; 82 31.15 4.325 15; 83 31.15 -1.2 15;
84 0 0 24.05; 85 0 4.325 24.05; 86 0 -1.2 24.05; 87 5.1 0 24.05;
88 5.1 4.325 24.05; 89 5.1 -1.2 24.05; 90 10.6 0 24.05; 91 10.6 4.325 24.05;
92 10.6 -1.2 24.05; 93 16.6 0 24.05; 94 16.6 4.325 24.05; 95 16.6 -1.2 24.05;
96 21.1 0 24.05; 97 21.1 4.325 24.05; 98 21.1 -1.2 24.05; 99 26.4 0 24.05;
100 26.4 4.325 24.05; 101 26.4 -1.2 24.05; 102 31.15 0 24.05;
103 31.15 4.325 24.05; 104 31.15 -1.2 24.05; 114 5.1 0 4.95; 115 10.6 0 4.95;
116 16.6 0 4.95; 117 21.1 0 4.95; 118 26.4 0 4.95; 119 27.1 0 0;
120 27.1 0 4.95;
MEMBER INCIDENCES
3 1 4; 31 1 23; 32 1 24; 33 4 25; 34 24 25; 35 24 26; 36 24 27; 37 25 28;
38 27 28; 39 27 29; 40 27 30; 41 28 31; 42 30 31; 43 30 32; 44 30 33; 45 31 34;
46 33 34; 47 33 35; 48 33 36; 49 34 37; 50 36 37; 51 36 38; 52 36 119;
53 37 40; 54 39 40; 55 39 41; 56 1 42; 57 4 43; 68 39 60; 69 40 61; 70 42 43;
71 42 44; 73 43 46; 74 46 114; 77 46 49; 78 49 115; 81 49 52; 82 52 116;
85 52 55; 86 55 117; 89 55 58; 90 58 118; 93 58 61; 94 60 61; 95 60 62;
96 42 63; 97 43 64; 108 60 81; 109 61 82; 110 63 64; 111 63 65; 113 64 67;
114 67 68; 117 67 70; 118 70 71; 121 70 73; 122 73 74; 125 73 76; 126 76 77;
129 76 79; 130 79 80; 133 79 82; 134 81 82; 135 81 83; 136 63 84; 137 64 85;
148 81 102; 149 82 103; 150 84 85; 151 84 86; 152 84 87; 153 85 88; 154 87 88;
155 87 89; 156 87 90; 157 88 91; 158 90 91; 159 90 92; 160 90 93; 161 91 94;
162 93 94; 163 93 95; 164 93 96; 165 94 97; 166 96 97; 167 96 98; 168 96 99;
169 97 100; 170 99 100; 171 99 101; 172 99 102; 173 100 103; 174 102 103;
175 102 104; 190 114 47; 191 24 114; 192 115 50; 193 27 115; 194 116 53;
195 30 116; 196 117 56; 197 33 117; 198 118 59; 199 119 120; 200 119 39;
201 42 114; 202 114 115; 203 115 116; 204 116 117; 205 117 120; 206 120 60;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
110 111 113 114 117 118 121 122 125 126 129 130 133 TO 135 150 151 154 155 -
158 159 162 163 166 167 170 171 174 175 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
32 33 36 37 40 41 44 45 48 49 52 53 56 57 68 69 152 156 160 164 168 172 191 -

193 195 197 199 TO 206 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
70 71 74 78 82 86 90 94 95 190 192 194 196 198 PRIS YD 0.55 ZD 0.3
MEMBER PROPERTY AMERICAN
97 109 137 149 PRIS YD 0.6 ZD 0.25
MEMBER PROPERTY AMERICAN
96 108 136 148 PRIS YD 0.7 ZD 0.25
73 77 81 85 89 93 PRIS YD 0.55 ZD 0.3
153 157 161 165 169 173 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
3 31 34 35 38 39 42 43 46 47 50 51 54 55 PRIS YD 0.5 ZD 0.3
CONSTANTS
BETA 90 MEMB 3 31 54 55 70 71 94 95 110 111 134 135 150 151 174 175
MATERIAL CONCRETE ALL
SUPPORTS
23 26 29 32 35 38 41 44 47 50 53 56 59 62 65 68 71 74 77 80 83 86 89 92 95 -
98 101 104 FIXED
****
MEMBER CRACKED
3 34 38 42 46 50 54 70 71 74 78 82 86 90 94 95 110 111 114 118 122 126 130 -
134 135 150 151 154 155 158 159 162 163 166 167 170 171 174 175 190 192 194 -
196 198 REDUCTION RIY 0.700000 RIZ 0.700000
33 37 41 45 49 53 57 69 73 77 81 85 89 93 96 97 108 109 113 117 121 125 129 -
133 137 149 153 157 161 165 169 173 200 TO 205 -
206 REDUCTION RIY 0.350000 RIZ 0.350000
MEMBER OFFSET
33 START 0.125 0 0
53 END -0.125 0 0
37 41 45 49 53 START 0.25 0 0
33 37 41 45 49 END -0.25 0 0
77 81 85 89 93 117 121 125 129 133 157 161 165 169 173 START 0.3 0 0
73 77 81 85 89 113 117 121 125 129 153 157 161 165 169 END -0.3 0 0
73 113 153 START 0.15 0 0
93 133 173 END -0.15 0 0
97 109 137 149 START 0 0 0.3
57 69 97 109 137 149 END 0 0 -0.3
57 69 START 0 0 0.25
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
153 157 161 165 169 173 UNI 30.45
73 77 81 85 89 93 UNI 40.47
32 36 40 44 48 52 56 68 96 108 136 148 152 156 160 164 168 172 191 193 195 -
197 199 200 UNI 12
113 117 121 125 129 133 UNI 66.52
33 37 41 45 49 53 UNI 14.4
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 35.000 M 25.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG 0 4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

```
1 4.500 M 35.000 M 25.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.420 0.550
!> END GENERATED DATA BLOCK
INT -1.17762 -1.17762 HEIG  0  4.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 35.000 M 25.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.875 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.50938 -1.50938 HEIG  0  4.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 25.000 M 35.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.867 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.61444 1.61444 HEIG  0  4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 25.000 M 35.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.866 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.27655 -1.27655 HEIG  0  4.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.500 M 15.000 M 20.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG  0  4.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
153 157 161 165 169 173 UNI GY -30.45
73 77 81 85 89 93 UNI GY -40.47
57 97 137 UMOM GZ 1
69 109 149 UMOM GZ -1
33 37 41 45 49 53 UMOM GX -1
153 157 161 165 169 173 UMOM GX 1
32 36 40 44 48 52 56 68 96 108 136 148 152 156 160 164 168 172 191 193 195 -
197 199 TO 206 UNI GY -12
113 117 121 125 129 133 UNI GY -66.52
33 37 41 45 49 53 UNI GY -14.4
LOAD 4 LOADTYPE Live  TITLE LL
```

MEMBER LOAD
57 97 137 UMOM GZ 0.18
69 109 149 UMOM GZ -0.18
33 37 41 45 49 53 UMOM GX -0.18
153 157 161 165 169 173 UMOM GX 0.18
153 157 161 165 169 173 UNI GY -2.26
73 77 81 85 89 93 UNI GY -3.75
113 117 121 125 129 133 UNI GY -4.77
33 37 41 45 49 53 73 77 81 85 89 93 UNI GY -1.24
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4.5
WIND LOAD -X 1 TYPE 2 YR 0 4.5
WIND LOAD -Z 1 TYPE 3 YR 0 4.5
WIND LOAD -Z -1 TYPE 3 YR 0 4.5
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 4.5
WIND LOAD -Z 1 TYPE 5 YR 0 4.5
WIND LOAD -X 1 TYPE 6 YR 0 4.5
WIND LOAD -X -1 TYPE 6 YR 0 4.5
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 4.5
WIND LOAD -X -1 TYPE 2 YR 0 4.5
WIND LOAD -Z 1 TYPE 3 YR 0 4.5
WIND LOAD -Z -1 TYPE 3 YR 0 4.5
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4.5
WIND LOAD -Z -1 TYPE 5 YR 0 4.5
WIND LOAD -X 1 TYPE 6 YR 0 4.5
WIND LOAD -X -1 TYPE 6 YR 0 4.5
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
3 33 34 37 38 41 42 45 46 49 50 53 54 57 69 70 73 77 81 85 89 93 94 97 109 -
110 113 117 121 125 129 133 134 137 149 150 153 154 157 158 161 162 165 166 -
169 170 173 174 TEMP 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)

3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 MEMB 3 31 TO 57 68 TO 71 73 74 77 78 81 82 85 86 89 90 93 TO 97 108 -
109 TO 111 113 114 117 118 121 122 125 126 129 130 133 TO 137 148 TO 175 190 -
191 TO 199

```
CLB 0.04 MEMB 3 31 TO 57 68 TO 71 73 74 77 78 81 82 85 86 89 90 93 TO 97 108 -
109 TO 111 113 114 117 118 121 122 125 126 129 130 133 TO 137 148 TO 175 190 -
191 TO 199
CLS 0.04 MEMB 3 32 TO 34 36 TO 38 40 TO 42 44 TO 46 48 TO 50 52 TO 54 56 57 -
68 TO 70 73 74 77 78 81 82 85 86 89 90 93 94 96 97 108 TO 110 113 114 117 -
118 121 122 125 126 129 130 133 134 136 137 148 TO 150 152 TO 154 -
156 TO 158 160 TO 162 164 TO 166 168 TO 170 172 TO 174 191 193 195 197 199 -
200
CLT 0.04 MEMB 3 31 TO 57 68 TO 71 73 74 77 78 81 82 85 86 89 90 93 TO 97 108 -
109 TO 111 113 114 117 118 121 122 125 126 129 130 133 TO 137 148 TO 175 190 -
191 TO 199
CLS 0.04 MEMB 31 35 39 43 47 51 55 71 95 111 135 151 155 159 163 167 171 175 -
190 192 194 196 198
MINSEC 10 MEMB 3 31 TO 57 68 TO 71 73 74 77 78 81 82 85 86 89 90 93 TO 97 -
108 TO 111 113 114 117 118 121 122 125 126 129 130 133 TO 137 148 TO 175 -
190 TO 200
MINMAIN 20 MEMB 70 71 73 74 77 78 81 82 85 86 89 90 93 TO 95 110 111 113 114 -
117 118 121 122 125 126 129 130 133 TO 135 150 151 153 TO 155 157 TO 159 -
161 TO 163 165 TO 167 169 TO 171 173 TO 175 190 192 194 196 198
MINMAIN 16 MEMB 3 31 33 TO 35 37 TO 39 41 TO 43 45 TO 47 49 TO 51 53 TO 55
SFACE 0.5 MEMB 33 37 41 45 49 53 57 69
EFACE 0.5 MEMB 33 37 41 45 49 53 57 69
SFACE 0.6 MEMB 97 109 113 117 121 125 129 133 137 149 153 157 161 165 169 173
EFACE 0.6 MEMB 97 109 113 117 121 125 129 133 137 149 153 157 161 165 169 173
SFACE 0.55 MEMB 73 77 81 85 89 93
EFACE 0.55 MEMB 73 77 81 85 89 93
DESIGN COLUMN 3 31 34 35 38 39 42 43 46 47 50 51 54 55 70 71 74 78 82 86 90 -
94 95 110 111 114 118 122 126 130 134 135 150 151 154 155 158 159 162 163 -
166 167 170 171 174 175 190 192 194 196 198
DESIGN BEAM 32 33 36 37 40 41 44 45 48 49 52 53 56 57 68 69 73 77 81 85 89 -
93 96 97 108 109 113 117 121 125 129 133 136 137 148 149 152 153 156 157 -
160 161 164 165 168 169 172 173 191 193 195 197 199 TO 206
END CONCRETE DESIGN
FINISH
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ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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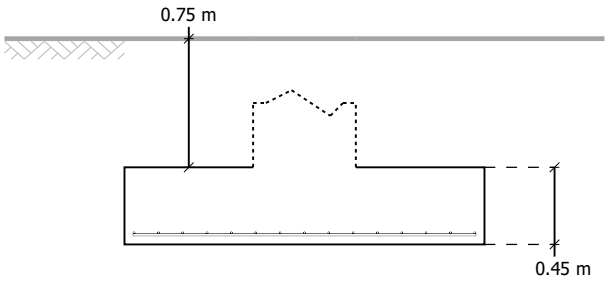
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Isolated Footing Design(ACI 318-08)

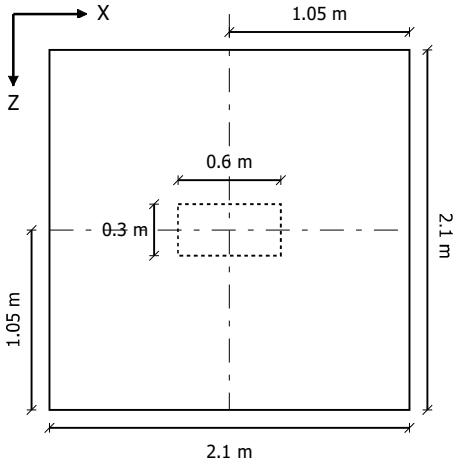
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
71	1	2.100m	2.100m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
71	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1-GRID-3 & C



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 2100.000mm
Footing Width - Z (Fw) : 2100.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.100m

Initial Width (W_o) = 2.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	392.294	-7.375	-0.007	-0.071	13.609
11	392.294	-7.375	-0.007	-0.071	13.609
12	417.182	-7.859	-0.007	-0.074	14.493
13	410.960	-7.738	-0.007	-0.073	14.272
14	392.299	-7.363	11.172	50.683	13.572
15	393.629	4.702	-0.001	-0.040	-23.528
16	412.040	0.521	-0.030	-0.167	-11.072
17	410.751	-7.898	-0.003	0.368	14.723
18	236.815	6.586	-0.035	-0.168	-25.626
19	235.098	-4.639	0.001	0.545	8.766

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	236.712	7.652	0.002	-0.012	-28.971
21	235.381	-4.413	11.175	50.711	8.128
22	393.733	3.636	-0.037	-0.196	-20.183
23	392.015	-7.589	-0.001	0.517	14.209
39	409.955	-15.923	-0.042	-0.228	39.407
40	410.738	-7.903	-0.006	-0.494	14.735
41	234.036	-15.339	-0.051	-0.250	41.678
42	235.080	-4.645	-0.002	-0.604	8.782
45	390.953	-18.289	-0.053	-0.278	47.122
46	391.997	-7.596	-0.005	-0.632	14.226

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	549.211	-10.325	-0.009	-0.099	19.052
25	561.656	-10.567	-0.010	-0.101	19.494
26	561.656	-10.567	-0.010	-0.101	19.494
27	511.726	-0.815	-0.033	-0.189	-9.287
28	482.751	-9.434	0.000	0.854	17.734
29	485.500	8.526	-0.057	-0.286	-37.294
30	482.751	-9.434	0.000	0.854	17.734
31	472.660	8.402	0.001	-0.042	-36.722
32	470.760	-8.833	15.962	72.420	16.278
33	355.367	10.980	-0.055	-0.264	-41.819
34	352.619	-6.980	0.002	0.877	13.209
35	353.072	-6.620	15.964	72.441	12.195
36	354.972	10.615	0.003	-0.020	-40.804
47	509.502	-18.355	-0.046	-0.255	44.556
48	482.723	-9.445	-0.005	-0.984	17.760
49	481.052	-26.555	-0.083	-0.418	70.394
50	482.723	-9.445	-0.005	-0.984	17.760
53	350.919	-24.100	-0.081	-0.395	65.869
54	352.590	-6.990	-0.003	-0.962	13.235

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.410m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.896m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.100 m Governing Load Case : # 10

Width (W_2) = 2.100 m Governing Load Case : # 10

Depth (D_2) = 0.450 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.410 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 47.626 kN

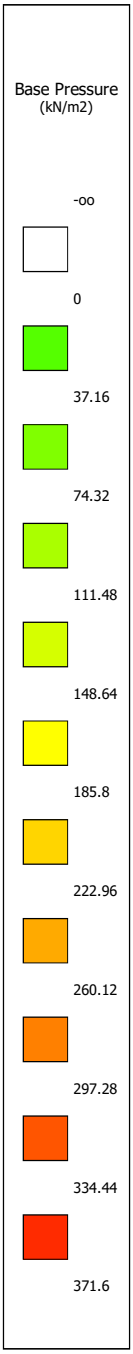
Gross Soil Bearing Capacity = 371.60 kN/m2

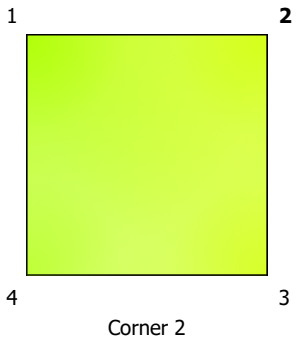
Soil Weight On Top Of Footing = 57.103 kN

Pressures at 4 Corners

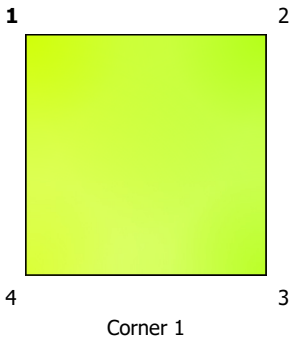
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

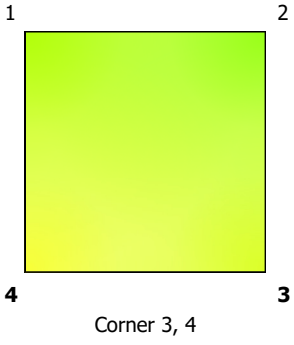




Corner 2

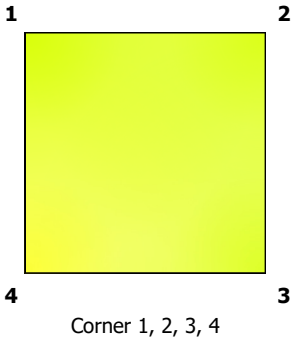


Corner 1



Corner 3, 4

Governing Base Pressure Contour



Corner 1, 2, 3, 4

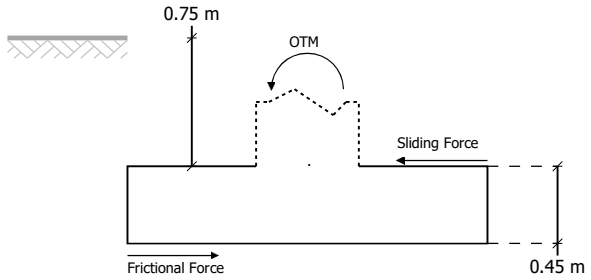
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**148.4574**	76.7334	76.3419	148.0659	0.000
15	96.4187	**129.6471**	129.5942	96.3658	0.000
14	87.5503	65.6712	**137.8593**	159.7384	0.000
14	87.5503	65.6712	137.8593	**159.7384**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**148.4574**	76.7334	76.3419	148.0659
15	96.4187	**129.6471**	129.5942	96.3658
14	87.5503	65.6712	**137.8593**	159.7384
14	87.5503	65.6712	137.8593	**159.7384**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	35.718	39858.098	50.513	1.500	7094.189	30.830	1.500
11	35.718	39858.098	50.513	1.500	7094.189	30.830	1.500
12	35.199	38763.064	49.779	1.500	7156.836	30.394	1.500
13	35.322	39022.766	49.953	1.500	7141.649	30.498	1.500
14	35.778	23.578	27.843	1.500	9.402	30.908	1.500
15	56.178	411417.427	79.448	1.500	12879.992	20.405	1.500
16	526.145	9139.944	742.852	1.500	3022.166	47.993	1.500
17	34.591	87762.550	48.919	1.500	1481.549	29.614	1.500
18	27.485	5234.630	38.870	1.500	1969.220	12.543	1.500
19	38.824	147267.698	54.906	1.500	657.045	32.876	1.500
20	23.650	90391.483	33.446	1.500	32062.657	11.060	1.500
21	40.850	16.131	21.218	1.500	6.441	35.309	1.500
22	72.659	7096.977	102.750	1.500	2470.613	23.987	1.500
23	34.691	185274.207	49.061	1.500	1013.652	29.594	1.500
39	17.131	6481.243	24.227	1.500	2193.058	11.604	1.500
40	34.569	49242.547	48.887	1.500	1094.511	29.590	1.500
41	11.705	3538.329	16.553	1.500	1311.990	7.322	1.500
42	38.769	88981.641	54.827	1.500	593.384	32.816	1.500
45	14.364	4920.892	20.314	1.500	1729.110	9.403	1.500
46	34.661	56397.834	49.017	1.500	825.602	29.561	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41

Governing Disturbing Force : -15.339kN

Governing Restoring Force : 179.545kN

Minimum Sliding Ratio for the Critical Load Case : 11.705

Critical Load Case for Overturning about X-Direction : 21

Governing Overturning Moment : 55.740kNm

Governing Resisting Moment : 359.023kNm

Minimum Overturning Ratio for the Critical Load Case : 6.441

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 21

Governing Disturbing Force : 11.175kN

Governing Restoring Force : 180.259kN

Minimum Sliding Ratio for the Critical Load Case : 16.131

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 48.581kNm

Governing Resisting Moment : 355.697kNm

Minimum Overturning Ratio for the Critical Load Case : 7.322

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 41

Governing Disturbing Force : 15.339kN

Governing Restoring Force : 253.916kN

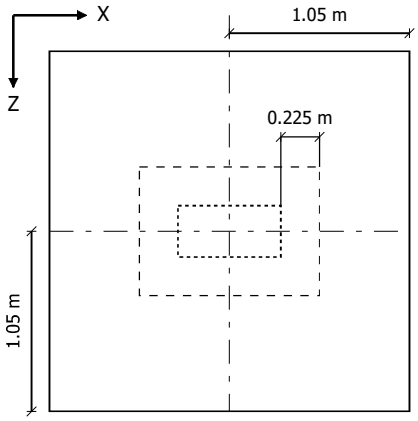
Minimum Sliding Ratio for the Critical Load Case : 16.553

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 562.145kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.376m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2613.679kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4357.164kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2613.679kN

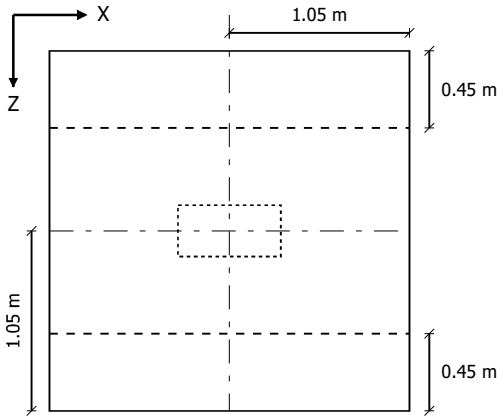
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1960.259kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 812.904kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.506m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

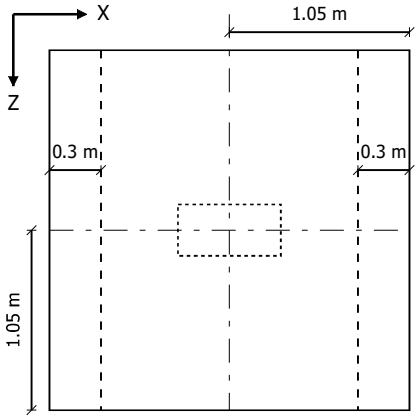
From above calculations, 0.75 X V_c = 609.678 kN

Critical load case for V_{ux} is # 32 $V_{ux} = V_{ux}|_{x=D_x}$ = 180.263 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 812.904 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.356 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

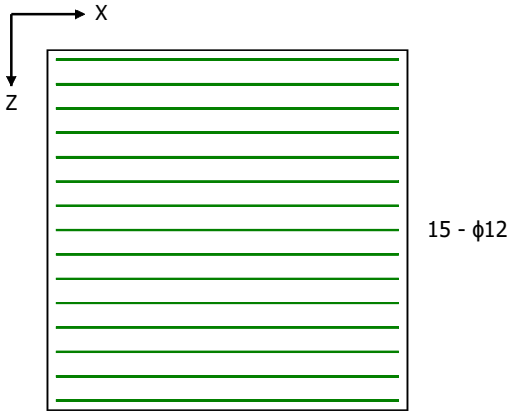
From above calculations, 0.75 X V_c = 609.678 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 132.427 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00163
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.750 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	102.452 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	113.836 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00084
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00073
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1537.378 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.700	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	15	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1696.448	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.394	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00205	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

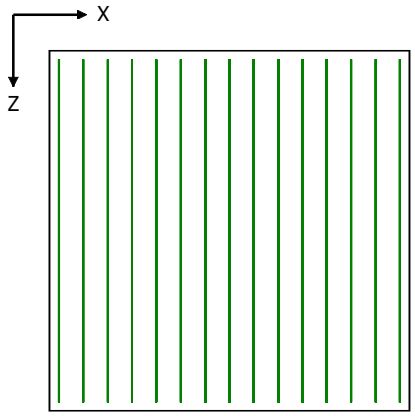
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.103

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 32

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.382	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00169

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.200 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 142.339 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 158.155 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00124

(Based on gross depth) ρ x d_{eff} / Depth = 0.00105

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1594.136 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 142.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 140mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.850 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.382 m

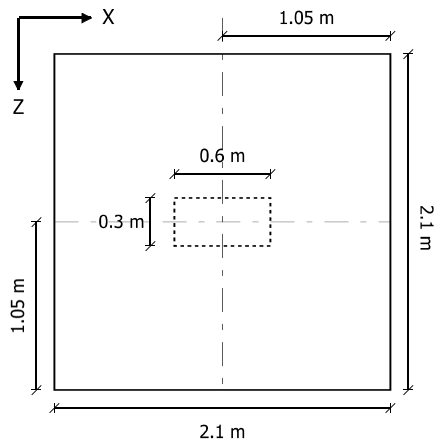
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00211

From ACI Cl.7.6.1, minimum req'd clear distance between bars

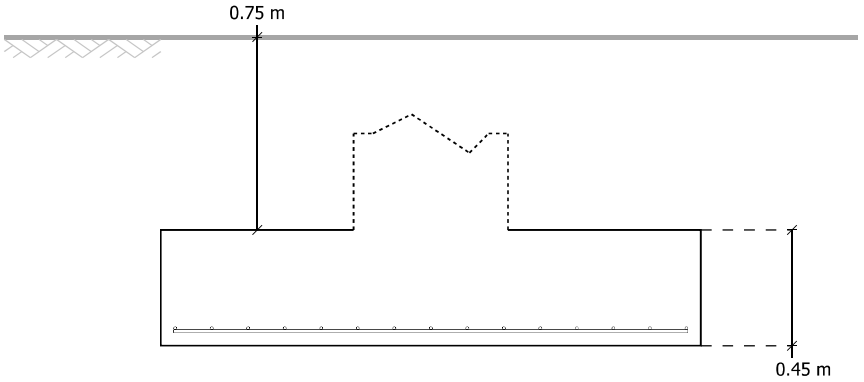
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.064

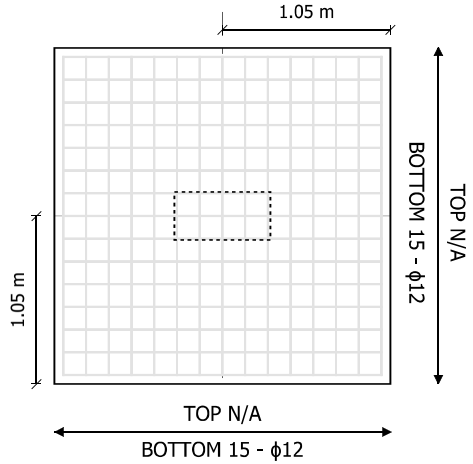
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	2.1 m	2.1 m	0.45 m	N/A	N/A	15 - ϕ12	15 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

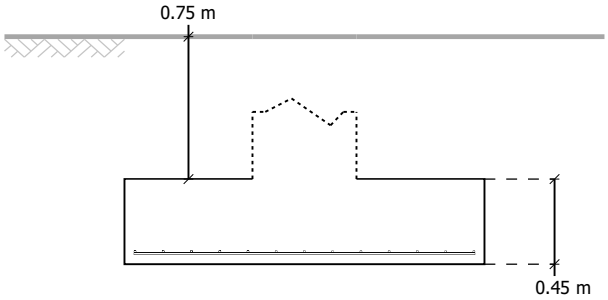
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

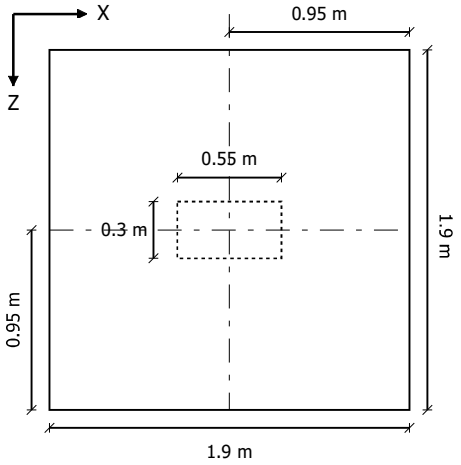
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
50	1	1.900m	1.900m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
50	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2-GRID-2 & C



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 1900.000mm
Footing Width - Z (Fw) : 1900.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.550m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.900m

Initial Width (W_o) = 1.900m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	374.933	1.558	23.629	8.514	-0.521
11	374.933	1.558	23.629	8.514	-0.521
12	400.840	2.149	23.347	8.364	-0.609
13	394.363	2.001	23.418	8.402	-0.587
14	382.306	1.452	22.761	13.769	-0.457
15	375.910	9.938	23.632	8.504	-17.184
16	384.801	13.395	-9.963	-12.677	-17.692
17	405.546	0.307	59.007	30.986	0.828
18	212.210	16.126	-30.329	-22.997	-23.120
19	239.870	-1.324	61.630	35.221	1.574

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	225.937	9.314	14.180	5.099	-16.976
21	232.333	0.829	13.310	10.363	-0.248
22	362.183	16.749	-20.878	-19.591	-23.329
23	389.843	-0.700	71.082	38.626	1.365
39	383.174	-8.642	-9.962	-12.656	15.929
40	385.323	0.561	-4.696	-9.461	0.653
41	210.041	-13.256	-30.328	-22.969	21.708
42	212.906	-0.986	-23.308	-18.709	1.340
45	360.014	-12.633	-20.877	-19.563	21.499
46	362.879	-0.363	-13.856	-15.303	1.131

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	524.906	2.182	33.081	11.920	-0.730
25	537.859	2.477	32.940	11.845	-0.773
26	537.859	2.477	32.940	11.845	-0.773
27	481.171	14.968	-7.702	-12.507	-19.011
28	486.729	-1.449	104.138	58.322	2.349
29	442.473	26.471	-42.997	-34.827	-37.161
30	486.729	-1.449	104.138	58.322	2.349
31	451.316	13.840	28.359	10.203	-24.430
32	460.453	1.719	27.115	17.724	-0.533
33	317.040	25.708	-49.945	-37.306	-36.961
34	361.296	-2.212	97.190	55.842	2.549
35	347.973	1.251	20.026	15.170	-0.377
36	338.836	13.373	21.270	7.649	-24.273
47	479.436	-8.538	-7.701	-12.485	16.851
48	443.587	-0.908	-31.762	-27.966	1.975
49	439.003	-20.541	-42.995	-34.782	34.563
50	443.587	-0.908	-31.762	-27.966	1.975
53	313.570	-21.304	-49.943	-37.261	34.763
54	318.153	-1.671	-38.710	-30.445	2.175

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.610m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.894m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.900 m Governing Load Case : # 10

Width (W_2) = 1.900 m Governing Load Case : # 10

Depth (D_2) = 0.450 m Governing Load Case : # 29

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.610 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 38.987 kN

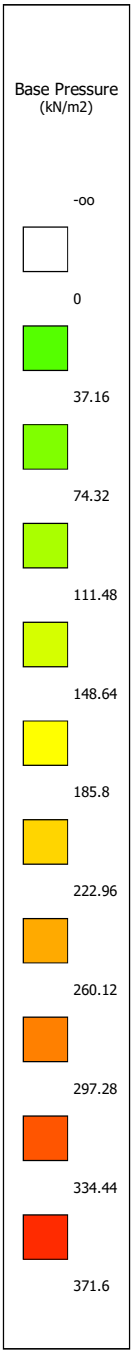
Gross Soil Bearing Capacity = 371.60 kN/m2

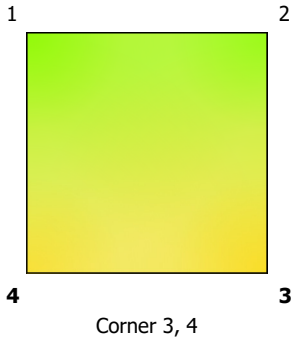
Soil Weight On Top Of Footing = 46.506 kN

Pressures at 4 Corners

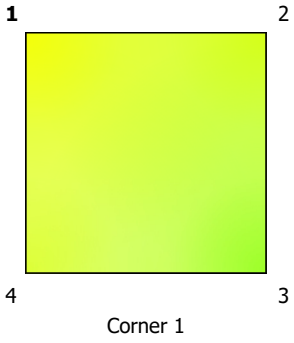
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

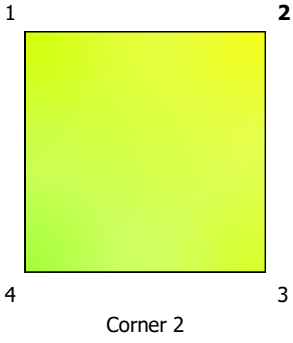




Corner 3, 4

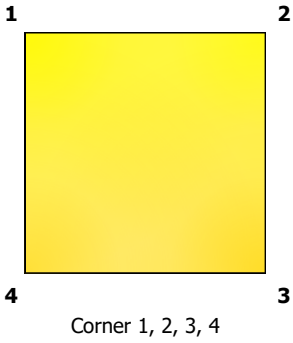


Corner 1



Corner 2

Governing Base Pressure Contour



Corner 1, 2, 3, 4

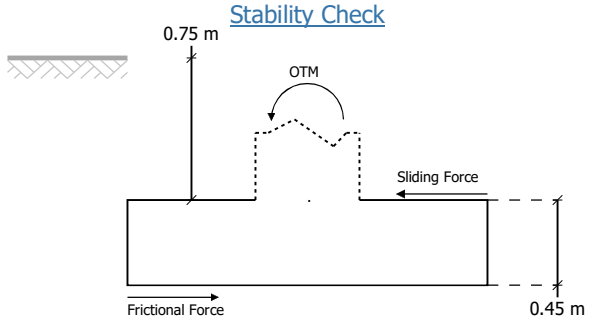
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**172.5206**	124.9609	74.2975	121.8573	0.000
22	122.3656	**176.3672**	125.6541	71.6525	0.000
23	71.3713	68.4317	**191.9724**	194.9121	0.000
23	71.3713	68.4317	191.9724	**194.9121**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**172.5206**	124.9609	74.2975	121.8573
22	122.3656	**176.3672**	125.6541	71.6525
23	71.3713	68.4317	**191.9724**	194.9121
23	71.3713	68.4317	191.9724	**194.9121**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	156.591	10.327	14.573	1.500	22.925	357.911	1.500
11	156.591	10.327	14.573	1.500	22.925	357.911	1.500
12	119.937	11.040	15.547	1.500	24.565	293.328	1.500
13	127.072	10.860	15.303	1.500	24.150	306.599	1.500
14	170.712	10.893	15.374	1.500	18.572	400.488	1.500
15	24.608	10.348	13.490	1.500	22.983	20.249	1.500
16	18.608	25.019	21.116	1.500	26.125	18.843	1.500
17	846.830	4.411	6.237	1.500	8.134	676.598	1.500
18	9.784	5.202	6.496	1.500	7.760	9.316	1.500
19	130.267	2.798	3.956	1.500	4.934	142.568	1.500
20	17.721	11.640	13.759	1.500	25.906	13.985	1.500
21	203.194	12.656	17.864	1.500	18.558	486.436	1.500
22	14.166	11.365	12.536	1.500	14.725	13.784	1.500
23	359.683	3.544	5.012	1.500	6.417	268.856	1.500
39	28.742	24.935	26.636	1.500	26.067	22.475	1.500
40	445.164	53.134	74.614	1.500	38.775	1117.116	1.500
41	11.816	5.165	6.692	1.500	7.709	10.152	1.500
42	160.374	6.785	9.587	1.500	9.762	159.025	1.500
45	18.690	11.310	13.684	1.500	14.668	15.575	1.500
46	655.017	17.150	24.246	1.500	19.848	329.139	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18

Governing Disturbing Force : 16.126kN

Governing Restoring Force : 157.782kN

Minimum Sliding Ratio for the Critical Load Case : 9.784

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 62.954kNm

Governing Resisting Moment : 310.628kNm

Minimum Overturning Ratio for the Critical Load Case : 4.934

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 61.630kN

Governing Restoring Force : 172.442kN

Minimum Sliding Ratio for the Critical Load Case : 2.798

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -30.377kNm

Governing Resisting Moment : 282.988kNm

Minimum Overturning Ratio for the Critical Load Case : 9.316

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 61.644kN

Governing Restoring Force : 243.870kN

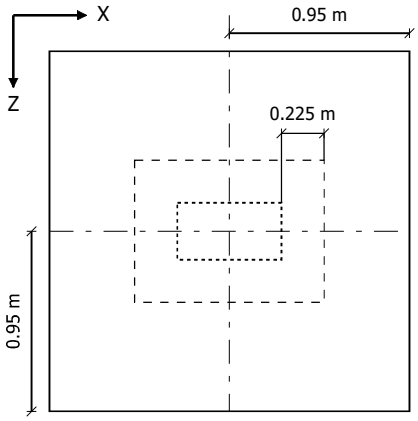
Minimum Sliding Ratio for the Critical Load Case : 3.956

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 1.833

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 510.227kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2651.544kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4318.454kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2536.260kN

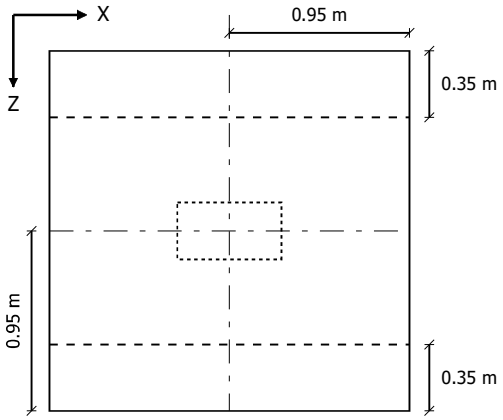
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1902.195kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 735.484kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.406m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

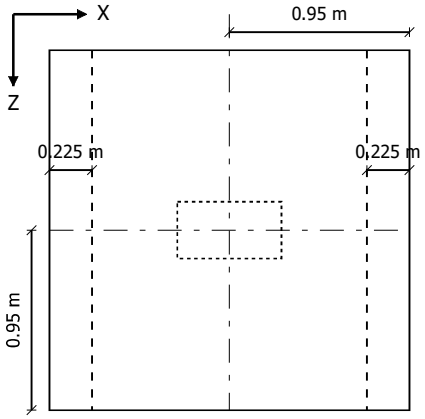
From above calculations, 0.75 X V_c = 551.613 kN

Critical load case for V_{ux} is # 28 $V_{ux} = V_{ux}|_{x=D_x}$ = 178.086 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 735.484 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.281 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

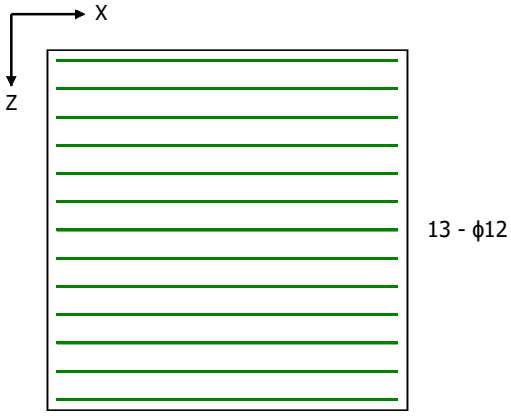
From above calculations, 0.75 X V_c = 551.613 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z} =$ 97.613 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00166
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.225 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	77.504 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	86.115 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00070
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00061
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1422.062 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 149.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.625	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	13	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1470.255	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.394	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00196	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

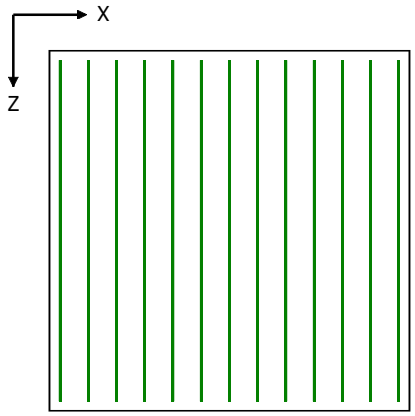
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.034

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.382	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.100 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 136.644 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 151.827 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00132

(Based on gross depth) ρ x d_{eff} / Depth = 0.00112

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1460.696 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 149.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.750 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.382 m

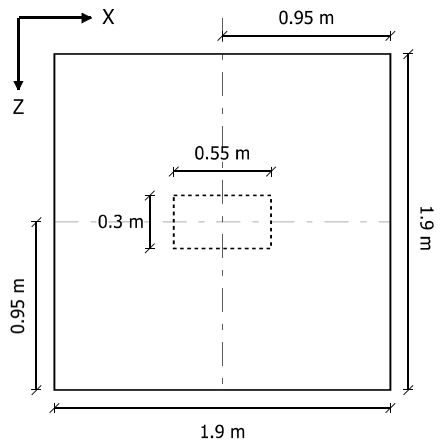
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00203

From ACI Cl.7.6.1, minimum req'd clear distance between bars

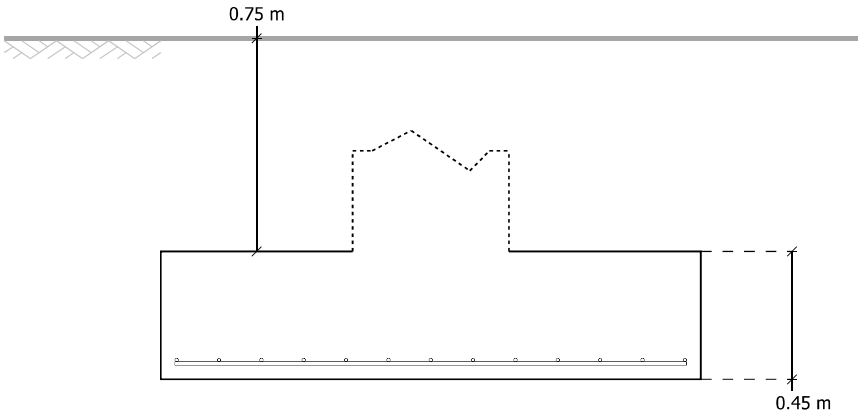
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.007

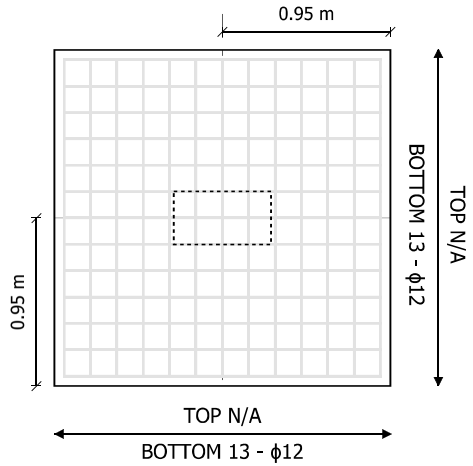
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.9 m	1.9 m	0.45 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

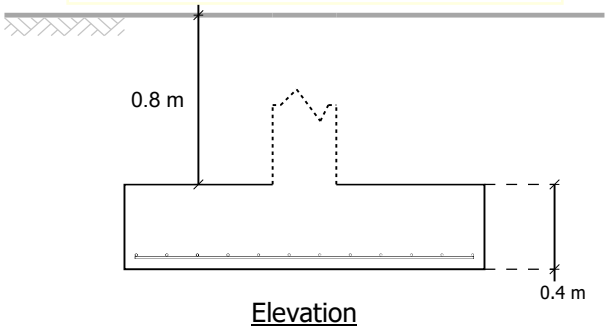
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

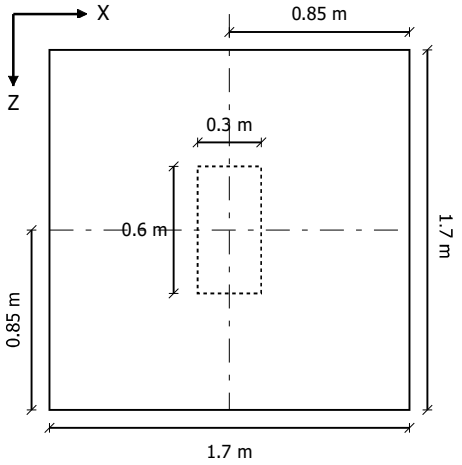
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
65	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
65	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F3-GRID-3 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	352.736	-11.834	5.257	-1.004	19.072
11	352.736	-11.834	5.257	-1.004	19.072
12	362.115	-12.548	5.168	-1.011	20.291
13	359.770	-12.369	5.190	-1.009	19.986
14	349.214	-11.593	16.237	15.186	18.869
15	348.185	-6.845	5.295	-0.957	8.187
16	356.114	12.919	5.310	1.180	-5.592
17	355.516	-32.922	13.359	13.340	37.132
18	206.767	26.617	3.314	2.316	-22.660
19	205.969	-34.504	14.046	18.529	34.305

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	207.090	-2.112	3.192	-0.556	0.559
21	208.120	-6.860	14.134	15.588	11.240
22	347.862	21.883	5.417	1.914	-15.032
23	347.064	-39.238	16.149	18.128	41.934
39	360.961	-31.130	5.232	1.084	40.104
40	362.998	-33.552	-2.995	-15.835	37.654
41	213.230	-32.115	3.210	2.188	38.268
42	215.945	-35.344	-7.759	-20.370	35.001
45	354.325	-36.849	5.313	1.787	45.897
46	357.040	-40.077	-5.656	-20.772	42.630

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	493.830	-16.568	7.360	-1.406	26.701
25	498.520	-16.925	7.315	-1.409	27.310
26	498.520	-16.925	7.315	-1.409	27.310
27	434.390	11.632	6.294	1.120	-2.447
28	418.897	-58.404	23.691	29.403	60.074
29	420.174	39.391	6.520	3.461	-31.070
30	418.897	-58.404	23.691	29.403	60.074
31	416.782	-7.074	6.362	-1.138	7.337
32	418.252	-13.857	21.993	21.924	22.596
33	309.664	43.297	4.987	3.766	-37.401
34	308.387	-54.497	22.158	29.707	53.743
35	312.431	-10.307	20.416	22.225	16.875
36	310.961	-3.524	4.785	-0.837	1.615
47	439.560	-35.354	6.211	1.018	46.296
48	434.858	-59.747	-11.197	-32.836	61.188
49	430.514	-54.582	6.354	3.257	66.415
50	434.858	-59.747	-11.197	-32.836	61.188
53	320.004	-50.675	4.821	3.562	60.084
54	324.348	-55.840	-12.730	-32.532	54.857

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.916m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 10

Width (W_2) = 1.700 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

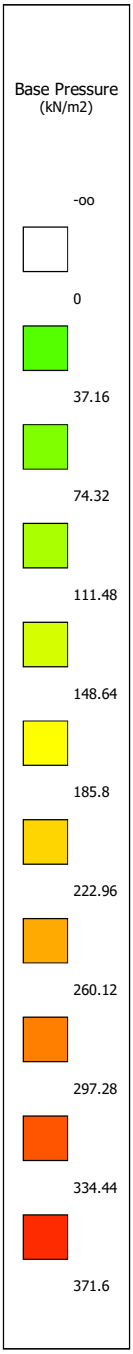
Gross Soil Bearing Capacity = 371.60 kN/m2

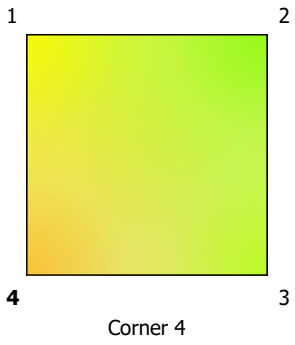
Soil Weight On Top Of Footing = 39.023 kN

Pressures at 4 Corners

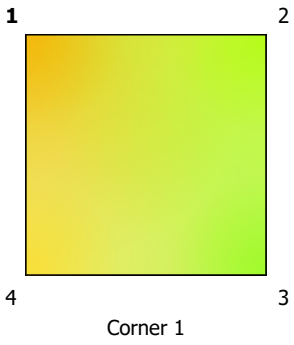
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

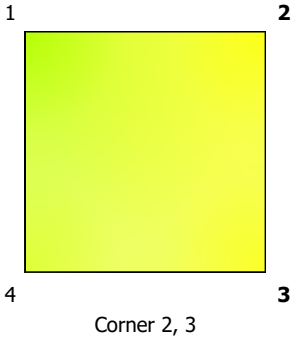




Corner 4

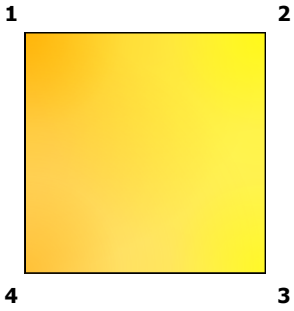


Corner 1



Corner 2, 3

Governing Base Pressure Contour



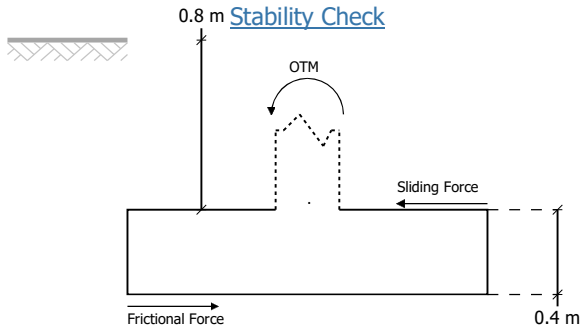
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**246.4172**	103.1363	46.8736	190.1545	0.000
22	109.4383	**167.5337**	177.5011	119.4057	0.000
22	109.4383	167.5337	**177.5011**	119.4057	0.000
23	183.5461	42.7859	102.8410	**243.6012**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**246.4172**	103.1363	46.8736	190.1545
22	109.4383	**167.5337**	177.5011	119.4057
22	109.4383	167.5337	**177.5011**	119.4057
23	183.5461	42.7859	102.8410	**243.6012**



Stability Check

-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	17.724	39.901	22.907	1.500	324.647	15.048	1.500
11	17.724	39.901	22.907	1.500	324.647	15.048	1.500
12	17.090	41.492	22.348	1.500	344.969	14.469	1.500
13	17.242	41.089	22.484	1.500	339.740	14.607	1.500
14	17.940	12.810	14.743	1.500	16.309	15.112	1.500
15	30.309	39.185	33.905	1.500	303.907	32.434	1.500
16	16.367	39.817	21.408	1.500	108.791	33.561	1.500
17	6.413	15.805	8.404	1.500	19.211	7.169	1.500
18	5.138	41.270	7.211	1.500	63.849	7.030	1.500
19	3.952	9.709	5.177	1.500	9.600	4.853	1.500
20	64.845	42.897	50.597	1.500	322.747	167.057	1.500
21	20.036	9.724	12.372	1.500	11.000	16.827	1.500
22	9.474	38.273	13.005	1.500	86.363	14.887	1.500
23	5.273	12.813	6.896	1.500	14.306	6.132	1.500
39	6.870	40.873	9.581	1.500	114.426	6.949	1.500
40	6.404	71.757	9.021	1.500	21.447	7.184	1.500
41	4.359	43.611	6.134	1.500	68.536	4.688	1.500
42	3.999	18.218	5.525	1.500	10.237	4.924	1.500
45	5.714	39.629	7.998	1.500	91.497	5.930	1.500
46	5.287	37.462	7.404	1.500	15.639	6.169	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 19

Governing Disturbing Force : -34.504kN

Governing Restoring Force : 136.368kN

Minimum Sliding Ratio for the Critical Load Case : 3.952

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 24.148kNm

Governing Resisting Moment : 231.821kNm

Minimum Overturning Ratio for the Critical Load Case : 9.600

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 14.046kN

Governing Restoring Force : 136.368kN

Minimum Sliding Ratio for the Critical Load Case : 9.709

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 51.114kNm

Governing Resisting Moment : 239.645kNm

Minimum Overturning Ratio for the Critical Load Case : 4.688

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 37.254kN

Governing Restoring Force : 192.853kN

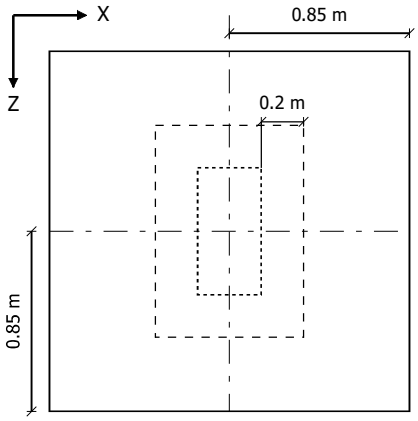
Minimum Sliding Ratio for the Critical Load Case : 5.177

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 446.373kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

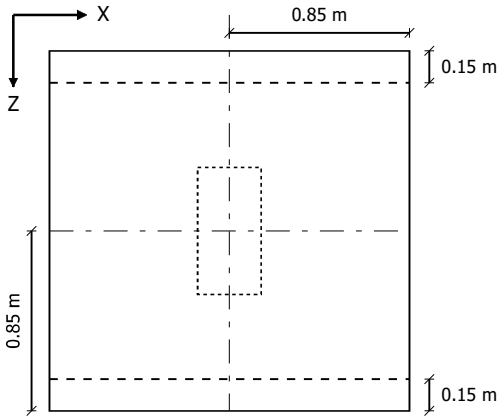
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.206m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

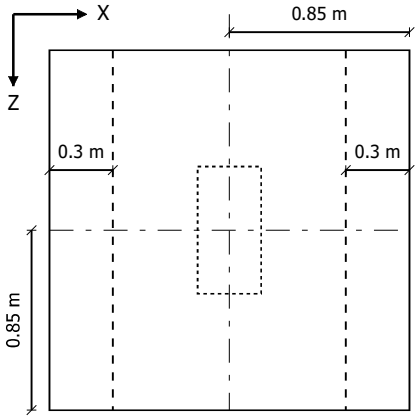
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 50 $V_{ux} = V_{ux}|_{x=D_x}$ = 74.810 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.356 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

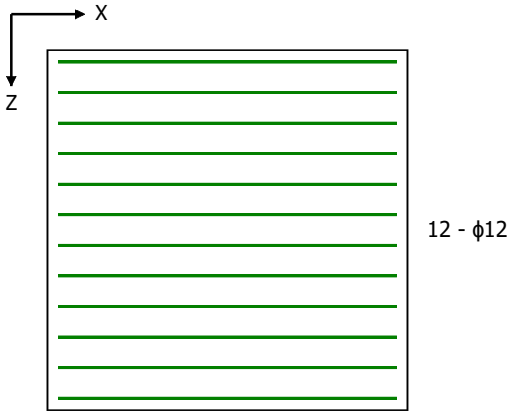
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z} =$ 155.702 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00172
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.700 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	104.221 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	115.801 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00138
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00119
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1166.812 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) - C_{cover}$ =	0.650	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	12	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1357.159	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00232	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

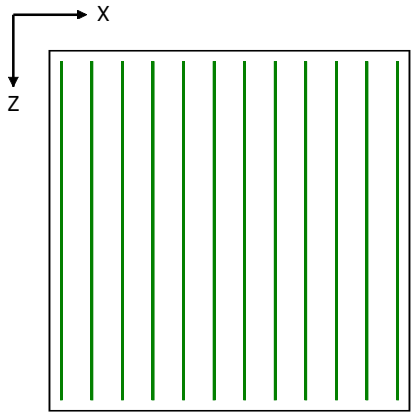
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.163

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	

From ACI Cl.7.12.2, ρ_{min} = 0.00167

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.550 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 53.817 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 59.797 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00076

(Based on gross depth) ρ x d_{eff} / Depth = 0.00063

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1137.580 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.500 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

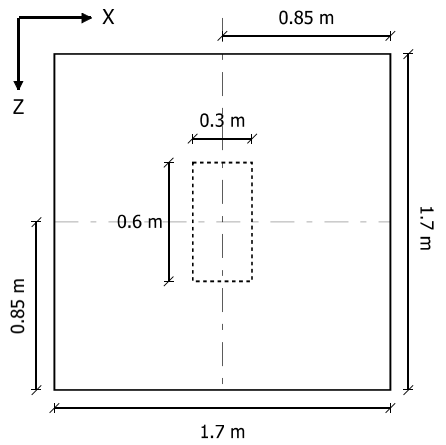
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

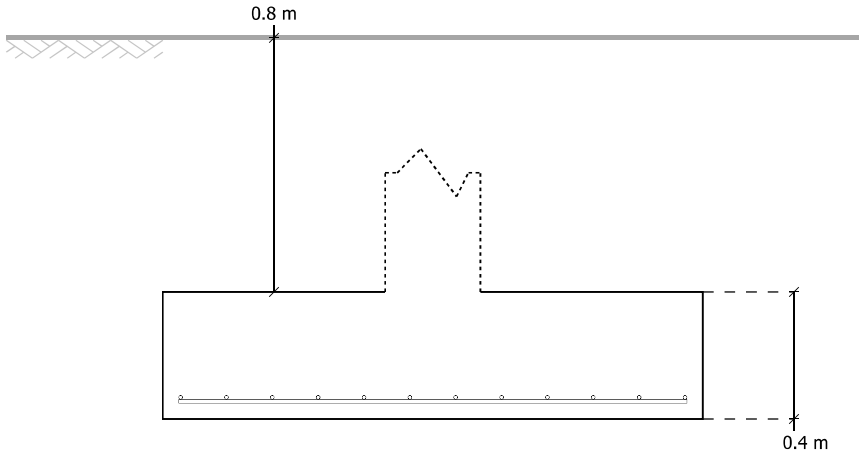
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.193

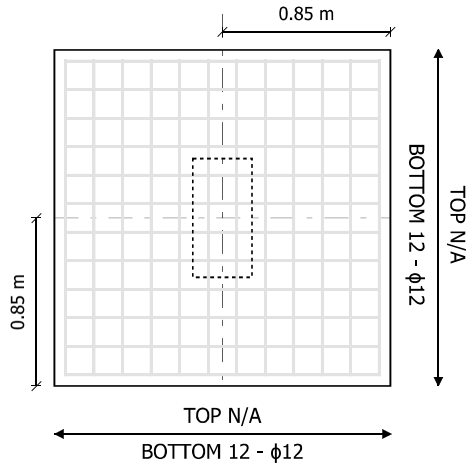
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

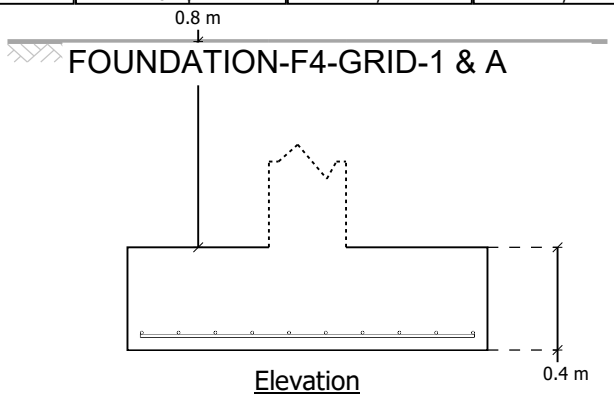
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

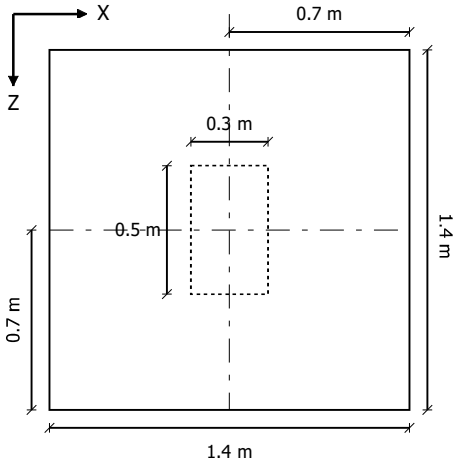
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
23	1	1.400m	1.400m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
23	10 - ϕ12	10 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1400.000mm
Footing Width - Z (Fw) : 1400.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.400m

Initial Width (W_o) = 1.400m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	126.927	-12.589	-3.162	6.344	3.477
11	126.927	-12.589	-3.162	6.344	3.477
12	129.584	-12.307	-3.229	6.422	3.446
13	128.919	-12.377	-3.212	6.402	3.454
14	117.288	-12.523	3.873	19.591	3.433
15	124.255	-10.404	-3.102	6.412	1.015
16	126.932	-9.705	-4.574	7.549	0.861
17	120.502	-13.742	6.345	19.099	4.519
18	73.506	-3.991	-3.713	5.336	-1.371
19	64.933	-9.373	10.846	20.735	3.507

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	73.485	-5.368	-1.837	3.874	-0.376
21	66.518	-7.487	5.137	17.054	2.042
22	124.277	-9.027	-4.978	7.873	0.020
23	115.704	-14.409	9.581	23.272	4.898
39	130.725	-14.935	-4.700	7.408	5.917
40	137.786	-13.912	-12.461	-6.544	4.625
41	78.563	-10.964	-3.881	5.147	5.370
42	87.978	-9.600	-14.229	-13.456	3.648
45	129.334	-15.999	-5.146	7.684	6.761
46	138.748	-14.636	-15.494	-10.918	5.039

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	177.697	-17.625	-4.427	8.881	4.868
25	179.026	-17.484	-4.460	8.920	4.852
26	179.026	-17.484	-4.460	8.920	4.852
27	154.444	-11.805	-5.354	8.961	1.357
28	135.684	-17.878	16.562	34.737	6.431
29	149.401	-9.266	-6.733	10.099	-1.374
30	135.684	-17.878	16.562	34.737	6.431
31	148.496	-11.985	-3.708	7.710	0.655
32	138.543	-15.012	6.255	26.538	4.110
33	109.995	-5.630	-5.751	8.156	-2.402
34	96.278	-14.242	17.544	32.795	5.403
35	100.465	-11.235	7.204	24.635	3.066
36	110.418	-8.208	-2.759	5.807	-0.389
47	158.489	-17.383	-5.488	8.810	6.750
48	172.555	-18.240	-23.558	-19.967	6.656
49	157.492	-20.422	-7.001	9.796	9.412
50	172.555	-18.240	-23.558	-19.967	6.656
53	118.086	-16.786	-6.019	7.854	8.384
54	133.149	-14.605	-22.576	-21.910	5.628

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.960m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.868m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.400 m Governing Load Case : # 10

Width (W_2) = 1.400 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 48

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.960 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 18.815 kN

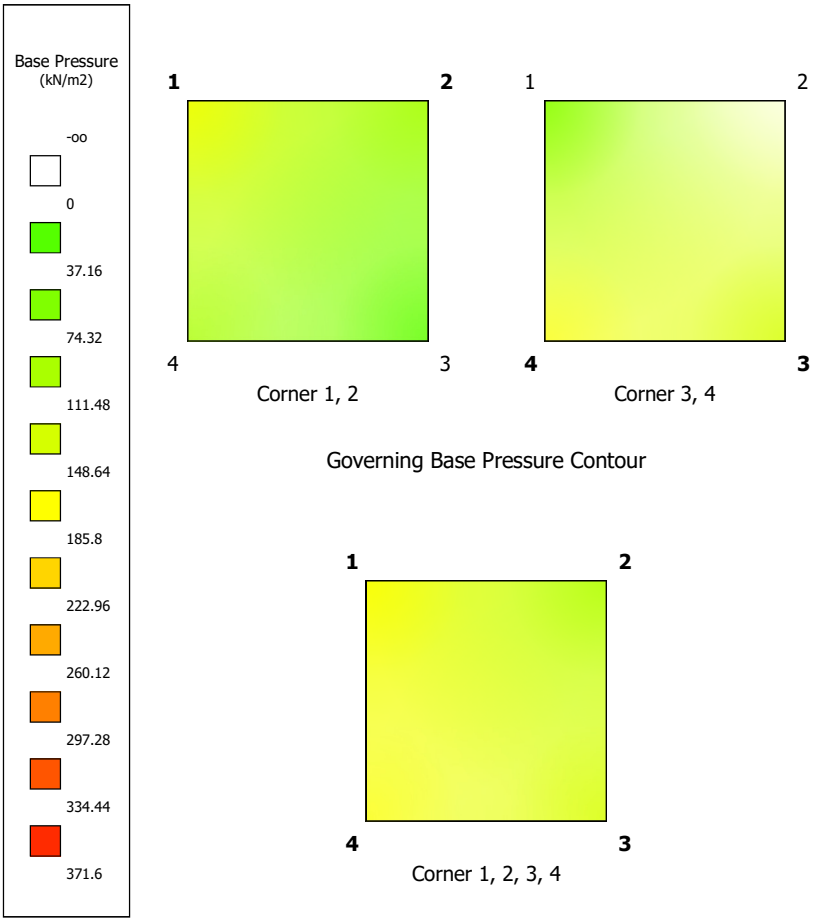
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 26.063 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



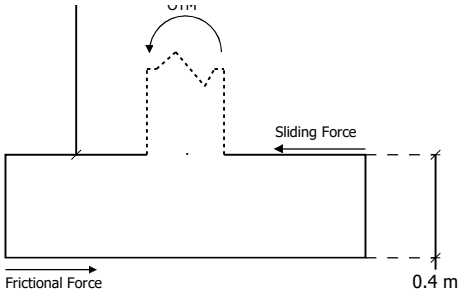
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**154.9320**	107.2931	32.4422	80.0811	0.000
46	154.9320	**107.2931**	32.4422	80.0811	0.000
23	45.9756	-0.6515	**117.8840**	164.5110	0.000
23	45.9756	-0.6515	117.8840	**164.5110**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**154.9320**	107.2931	32.4422	80.0811
46	154.9320	**107.2931**	32.4422	80.0811
23	45.9756	0.0000	**117.8841**	164.5113
23	45.9756	0.0000	117.8841	**164.5113**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	6.823	27.168	9.359	1.500	23.729	14.251	1.500
11	6.823	27.168	9.359	1.500	23.729	14.251	1.500
12	7.088	27.017	9.696	1.500	23.852	14.719	1.500
13	7.021	27.054	9.611	1.500	23.822	14.601	1.500
14	6.475	20.937	8.748	1.500	5.382	13.572	1.500
15	8.128	27.266	11.016	1.500	22.943	23.077	1.500
16	8.851	18.782	11.323	1.500	21.070	25.579	1.500
17	6.017	13.031	7.726	1.500	5.362	11.663	1.500
18	14.832	15.942	15.357	1.500	21.587	371.912	1.500
19	5.858	5.062	5.417	1.500	3.076	10.738	1.500
20	11.024	32.221	14.751	1.500	26.468	47.383	1.500
21	7.440	10.842	8.675	1.500	4.094	15.691	1.500
22	9.370	16.992	11.603	1.500	20.173	32.903	1.500
23	5.572	8.380	6.562	1.500	4.156	10.642	1.500
39	5.879	18.682	7.931	1.500	22.282	10.427	1.500
40	6.565	7.330	6.916	1.500	11.113	12.652	1.500
41	5.630	15.904	7.505	1.500	24.109	8.966	1.500
42	6.920	4.669	5.473	1.500	4.870	12.561	1.500
45	5.444	16.929	7.330	1.500	21.720	9.346	1.500
46	6.273	5.926	6.092	1.500	7.525	11.897	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 45
Governing Disturbing Force : -15.999kN
Governing Restoring Force : 87.106kN
Minimum Sliding Ratio for the Critical Load Case : 5.444
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 25.073kNm
Governing Resisting Moment : 77.119kNm
Minimum Overturning Ratio for the Critical Load Case : 3.076

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 42
Governing Disturbing Force : -14.229kN
Governing Restoring Force : 66.428kN

Minimum Sliding Ratio for the Critical Load Case : 4.669

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 9.756kNm

Governing Resisting Moment : 87.466kNm

Minimum Overturning Ratio for the Critical Load Case : 8.966

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 14.335kN

Governing Restoring Force : 77.649kN

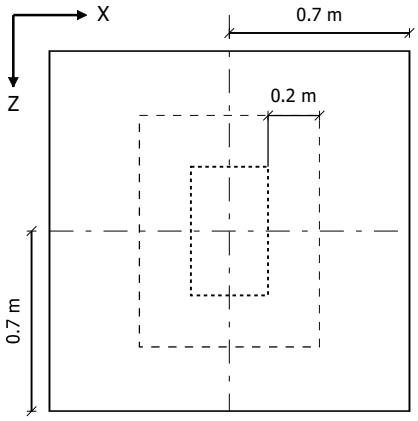
Minimum Sliding Ratio for the Critical Load Case : 5.417

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.667

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 161.812kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.976m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2212.777kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3331.062kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2011.615kN

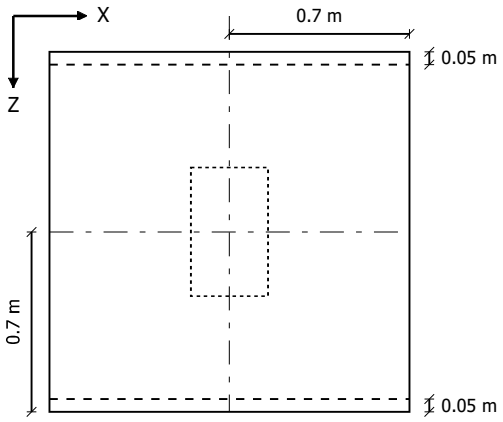
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1508.712kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 473.162kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.106m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

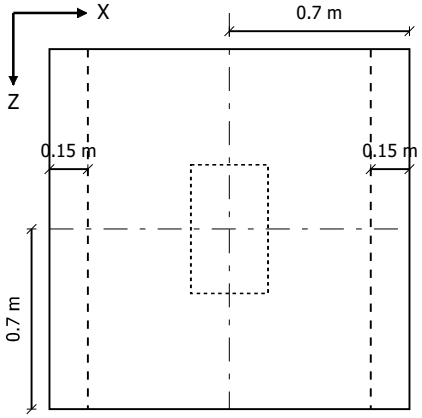
From above calculations, 0.75 X V_c = 354.872 kN

Critical load case for V_{ux} is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 25.419 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 473.162 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.206 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

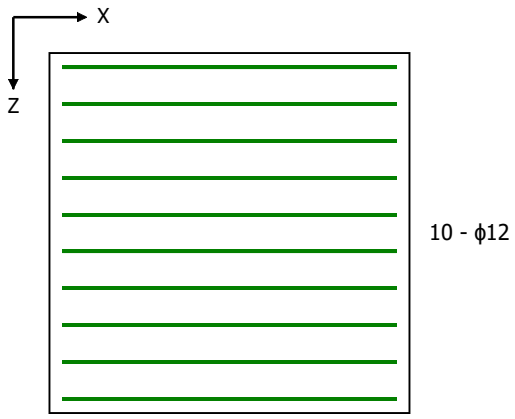
From above calculations, 0.75 X V_c = 354.872 kN

Critical load case for V_{uz} is # 48 $V_{uz} = V_{uz}|_{z=D_z}$ = 39.498 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00174	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.550	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	28.257	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	31.396	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00045	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00039	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	976.480	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover} =$ 0.500 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar} = \frac{A_s}{A_{bar}} =$ 10

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1130.965 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho = \frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00235

From ACI Cl.7.6.1, minimum req'd clear distance between bars

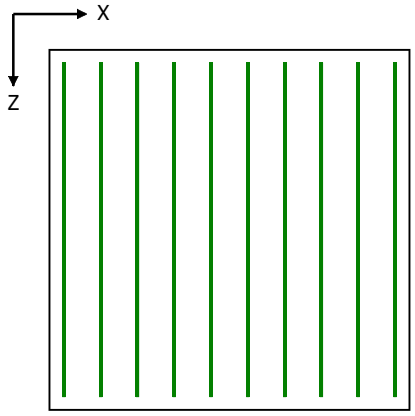
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.158

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



10 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal} = 0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max} = 0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00172

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.950 m

Ultimate moment, $M_u|_{x=D_x}$ = 23.743 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 26.382 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00041

(Based on gross depth) ρ x d_{eff} / Depth = 0.00034

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 964.080 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.400 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 10

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1130.965 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

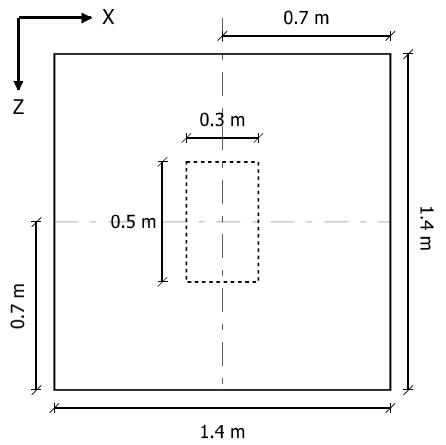
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00243

From ACI Cl.7.6.1, minimum req'd clear distance between bars

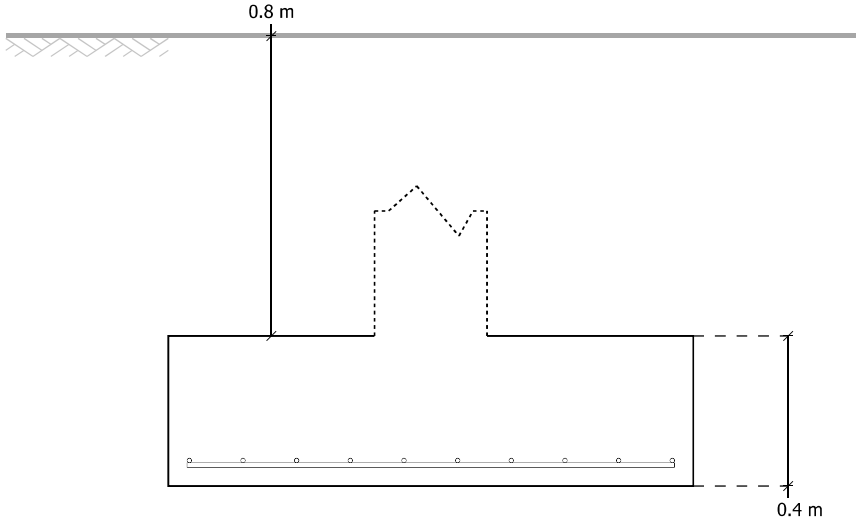
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.173

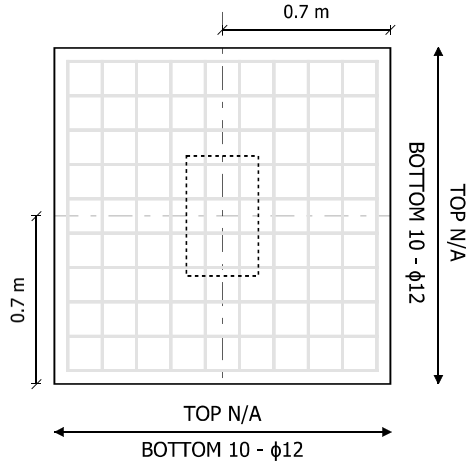
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.4 m	1.4 m	0.4 m	N/A	N/A	10 - ϕ12	10 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

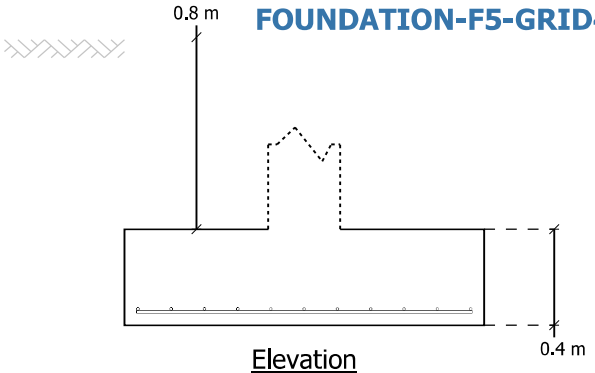
Print Calculation Sheet

Isolated Footing Design(ACI 318-05)

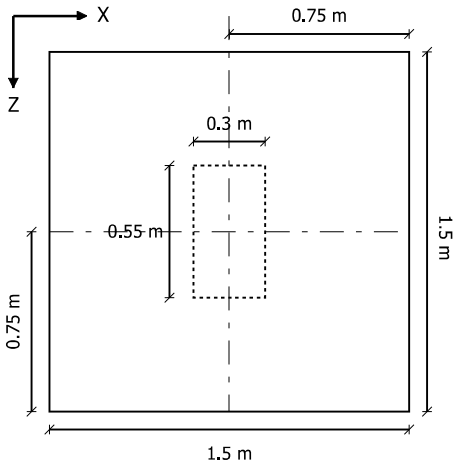
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
86	1	1.500m	1.500m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
86	11 - ϕ12	11 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F5-GRID4 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1500.000mm
Footing Width - Z (Fw) : 1500.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.550m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.500m

Initial Width (W_o) = 1.500m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	209.044	-12.474	53.529	5.218	4.291
11	209.044	-12.474	53.529	5.218	4.291
12	213.609	-12.063	53.573	5.185	4.216
13	212.467	-12.166	53.562	5.193	4.235
14	212.576	-12.501	57.758	17.480	4.319
15	205.529	-9.271	53.532	5.230	-0.068
16	211.320	-8.961	54.869	6.607	0.843
17	214.758	-13.852	57.903	13.823	5.613
18	123.896	-3.211	33.860	5.016	-1.947
19	128.480	-9.733	37.904	14.638	4.413

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	121.911	-4.282	32.120	3.143	-1.784
21	128.958	-7.512	36.346	15.393	2.602
22	207.514	-8.201	55.272	7.103	-0.231
23	212.098	-14.723	59.316	16.725	6.130
39	214.645	-14.955	54.860	6.579	7.293
40	210.256	-13.802	48.931	-3.753	5.568
41	128.329	-11.204	33.848	4.979	6.652
42	122.477	-9.667	25.942	-8.798	4.353
45	211.946	-16.194	55.260	7.066	8.369
46	206.095	-14.657	47.353	-6.711	6.069

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	292.661	-17.464	74.941	7.305	6.008
25	294.944	-17.258	74.963	7.288	5.970
26	294.944	-17.258	74.963	7.288	5.970
27	256.933	-10.892	65.700	7.717	1.411
28	258.021	-18.361	73.516	24.656	8.053
29	250.687	-7.926	67.045	9.262	-2.124
30	258.021	-18.361	73.516	24.656	8.053
31	245.831	-10.394	64.239	6.279	-1.078
32	255.898	-15.008	70.276	23.779	5.189
33	185.691	-4.390	50.964	7.713	-3.373
34	193.026	-14.824	57.435	23.107	6.803
35	193.185	-11.266	54.218	22.214	3.901
36	183.118	-6.651	48.180	4.713	-2.365
47	260.479	-17.286	65.691	7.687	8.291
48	248.417	-18.255	54.376	-12.841	7.956
49	257.780	-20.714	67.026	9.202	11.636
50	248.417	-18.255	54.376	-12.841	7.956
53	192.784	-17.178	50.946	7.653	10.386
54	183.421	-14.719	38.295	-14.390	6.707

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.250m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.274m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.500 m Governing Load Case : # 10

Width (W_2) = 1.500 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 30

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.250 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 21.599 kN

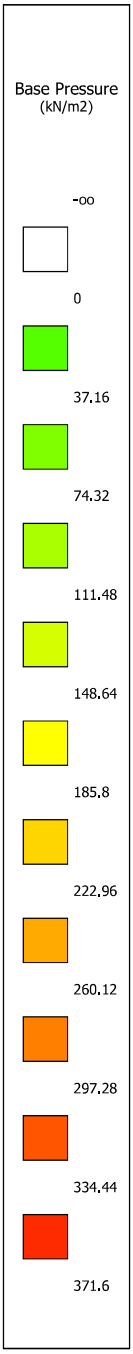
Gross Soil Bearing Capacity = 371.60 kN/m2

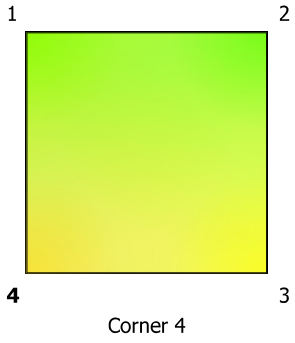
Soil Weight On Top Of Footing = 30.023 kN

Pressures at 4 Corners

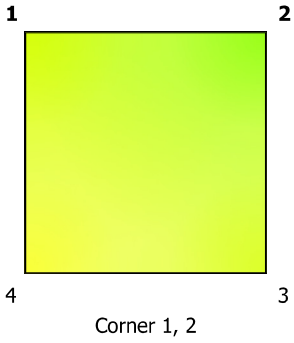
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

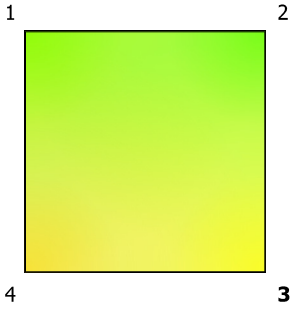




Corner 4

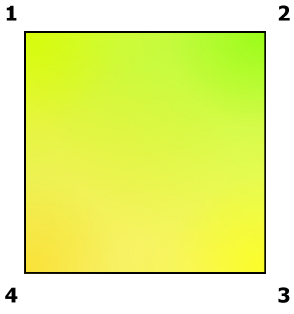


Corner 1, 2



Corner 3

Governing Base Pressure Contour



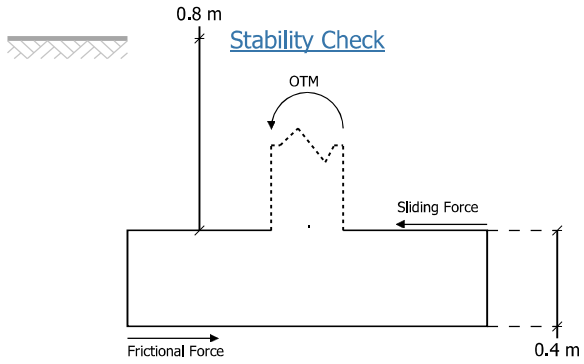
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**114.0101**	71.5856	115.0716	157.4961	0.000
46	114.0101	**71.5856**	115.0716	157.4961	0.000
14	61.8399	28.7050	**173.0026**	206.1375	0.000
23	66.6621	23.9288	167.7556	**210.4888**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**114.0101**	71.5856	115.0716	157.4961
46	114.0101	**71.5856**	115.0716	157.4961
14	61.8399	28.7050	**173.0026**	206.1375
23	66.6621	23.9288	167.7556	**210.4888**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	10.448	2.435	3.353	1.500	7.347	21.204	1.500
11	10.448	2.435	3.353	1.500	7.347	21.204	1.500
12	10.994	2.475	3.415	1.500	7.480	22.146	1.500
13	10.854	2.465	3.400	1.500	7.447	21.906	1.500
14	10.567	2.287	3.161	1.500	4.886	21.401	1.500
15	13.868	2.402	3.347	1.500	7.244	53.331	1.500
16	14.672	2.396	3.344	1.500	6.911	44.836	1.500
17	9.615	2.300	3.164	1.500	5.406	18.027	1.500
18	27.327	2.592	3.649	1.500	7.100	200.531	1.500
19	9.252	2.376	3.254	1.500	4.538	16.418	1.500
20	20.264	2.701	3.787	1.500	8.148	1831.856	1.500
21	12.020	2.484	3.440	1.500	4.530	24.387	1.500
22	15.799	2.344	3.279	1.500	6.658	64.158	1.500
23	8.956	2.223	3.051	1.500	4.893	16.565	1.500
39	8.902	2.427	3.311	1.500	7.006	15.141	1.500
40	9.487	2.676	3.642	1.500	12.425	17.829	1.500
41	8.031	2.658	3.569	1.500	7.296	12.238	1.500
42	9.005	3.356	4.447	1.500	82.801	16.044	1.500
45	8.138	2.385	3.237	1.500	6.782	13.402	1.500
46	8.792	2.721	3.676	1.500	15.816	16.308	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41

Governing Disturbing Force : -11.204kN

Governing Restoring Force : 89.976kN

Minimum Sliding Ratio for the Critical Load Case : 8.031

Critical Load Case for Overturning about X-Direction :	21
Governing Overturning Moment :	29.931kNm
Governing Resisting Moment :	135.581kNm
Minimum Overturning Ratio for the Critical Load Case :	4.530

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction :	23
Governing Disturbing Force :	59.316kN
Governing Restoring Force :	131.860kN
Minimum Sliding Ratio for the Critical Load Case :	2.223
Critical Load Case for Overturning about Z-Direction :	41
Governing Overturning Moment :	11.134kNm
Governing Resisting Moment :	136.257kNm
Minimum Overturning Ratio for the Critical Load Case :	12.238

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

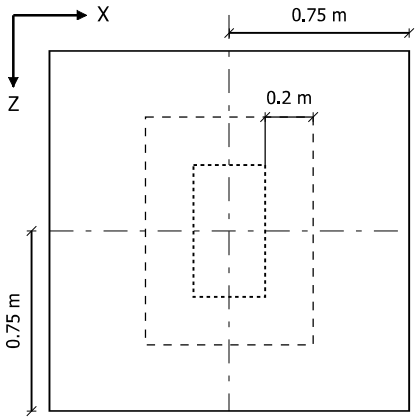
Critical Load Case for Sliding along Resultant Direction :	23
Governing Disturbing Force :	61.116kN
Governing Restoring Force :	186.478kN
Minimum Sliding Ratio for the Critical Load Case :	3.051

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.833

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 257.886kN, Load Case # 25

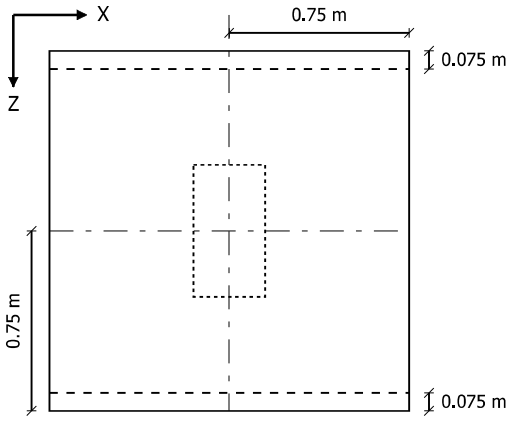
From ACI Cl.11.12.2.1, b_o for column=	$2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ =	3.076m
Equation 11-33, V_{c1} =	$0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$	2173.720kN
Equation 11-34, V_{c2} =	$0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$	3364.859kN
Equation 11-35, V_{c3} =	$0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$	2079.210kN
Punching shear strength, V_c =	0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) =	1559.408kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.3.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.131m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

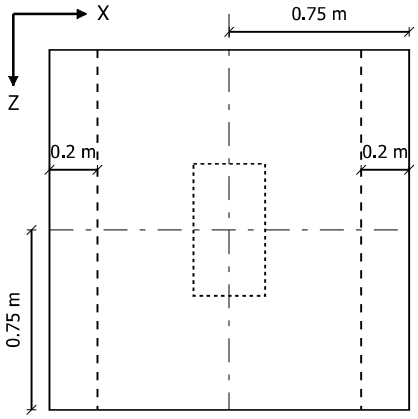
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{ux} is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 44.279 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.3.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

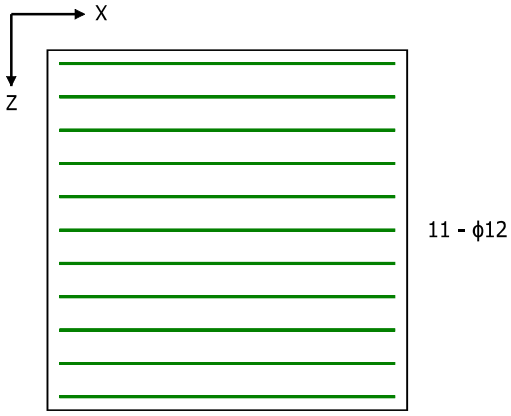
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{uz} is # 25 $V_{uz} = V_{uz}|_{z=D_z} =$ 66.436 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00174
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.600 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	46.117 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	51.241 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00069
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00059
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1042.370 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.550	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	11	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1244.062	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00241	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

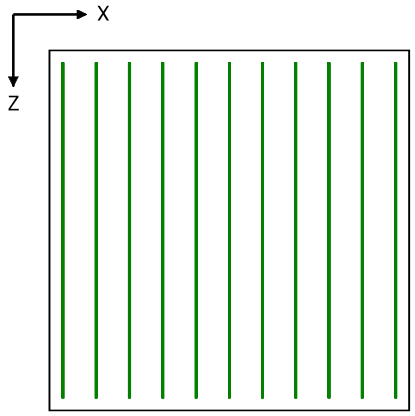
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.193

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



11 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 30

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.025 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 36.145 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 40.161 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00058

(Based on gross depth) ρ x d_{eff} / Depth = 0.00048

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1023.991 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.425 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

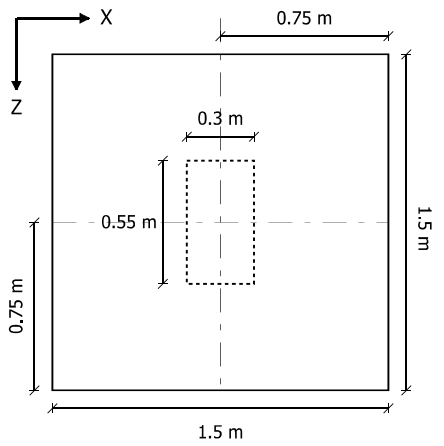
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00250

From ACI Cl.7.6.1, minimum req'd clear distance between bars

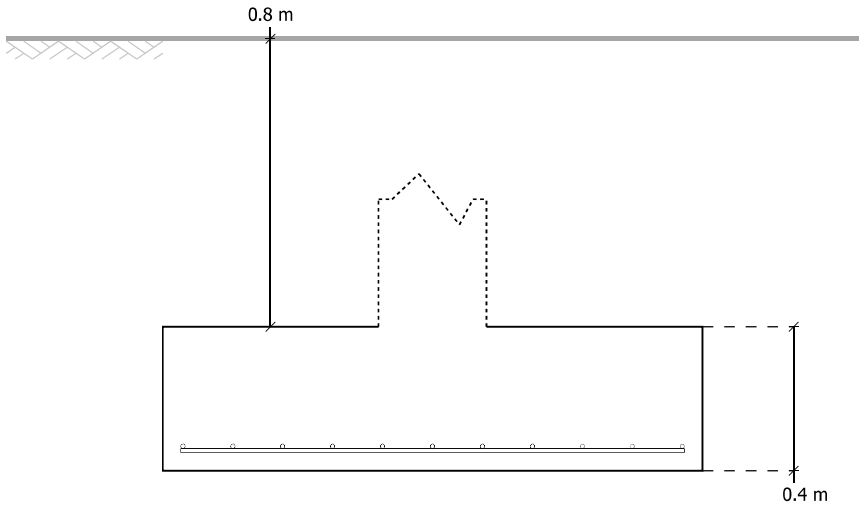
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.215

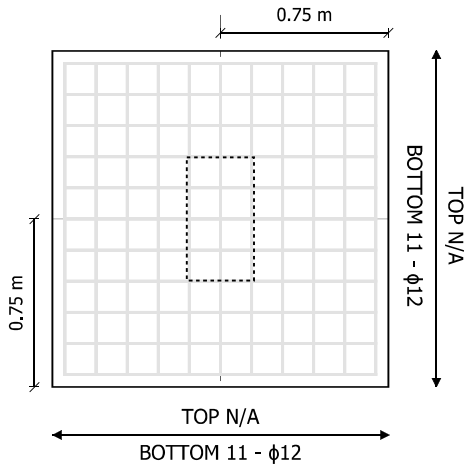
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.5 m	1.5 m	0.4 m	N/A	N/A	11 - ϕ12	11 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet